

Biscayne Bay Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

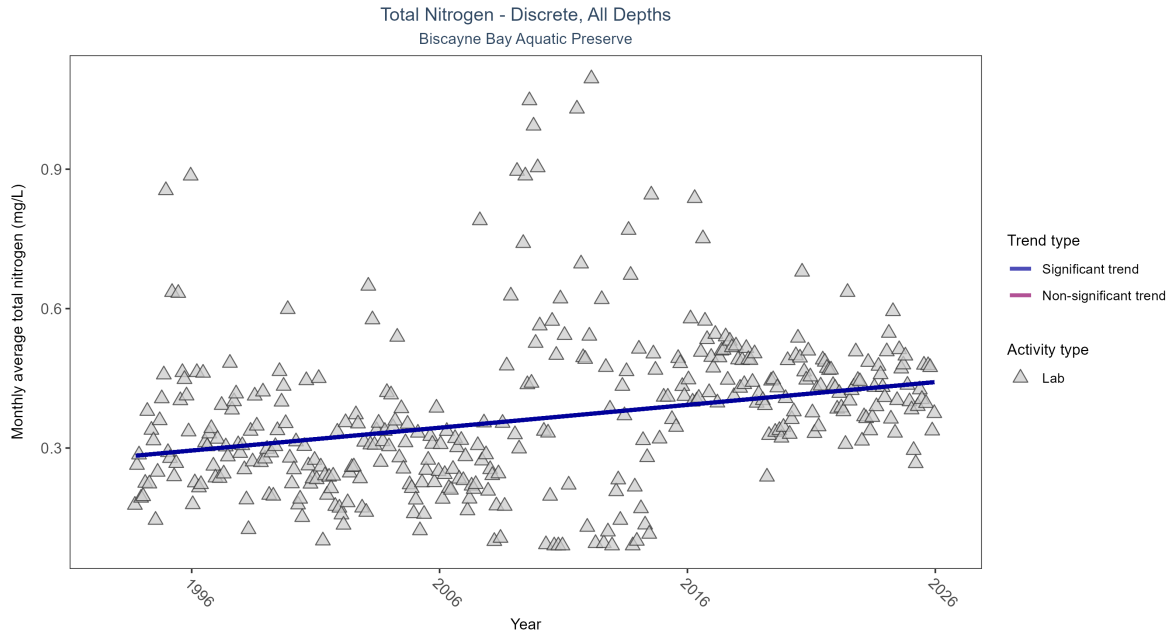


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	9435	33	1993 - 2025	0.335	0.24083	0.27988	0.00491	0

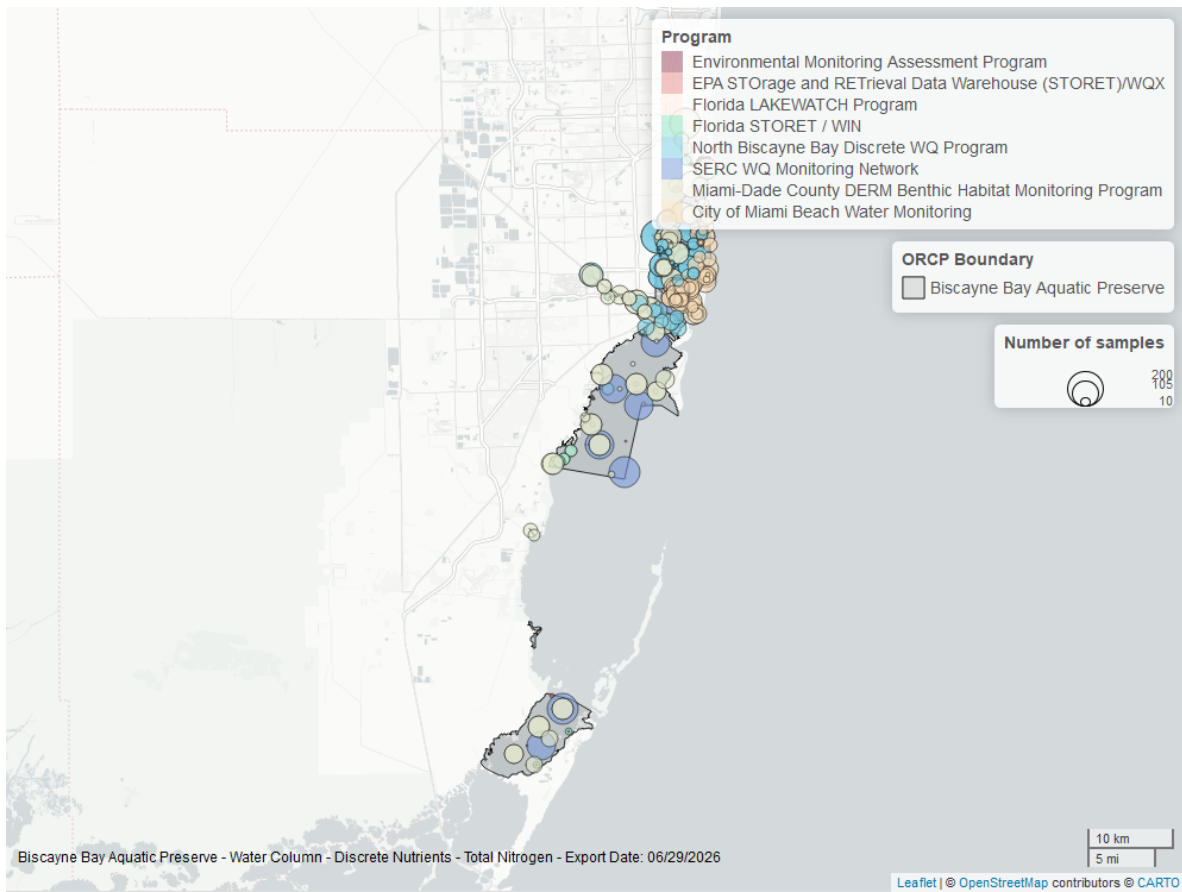


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

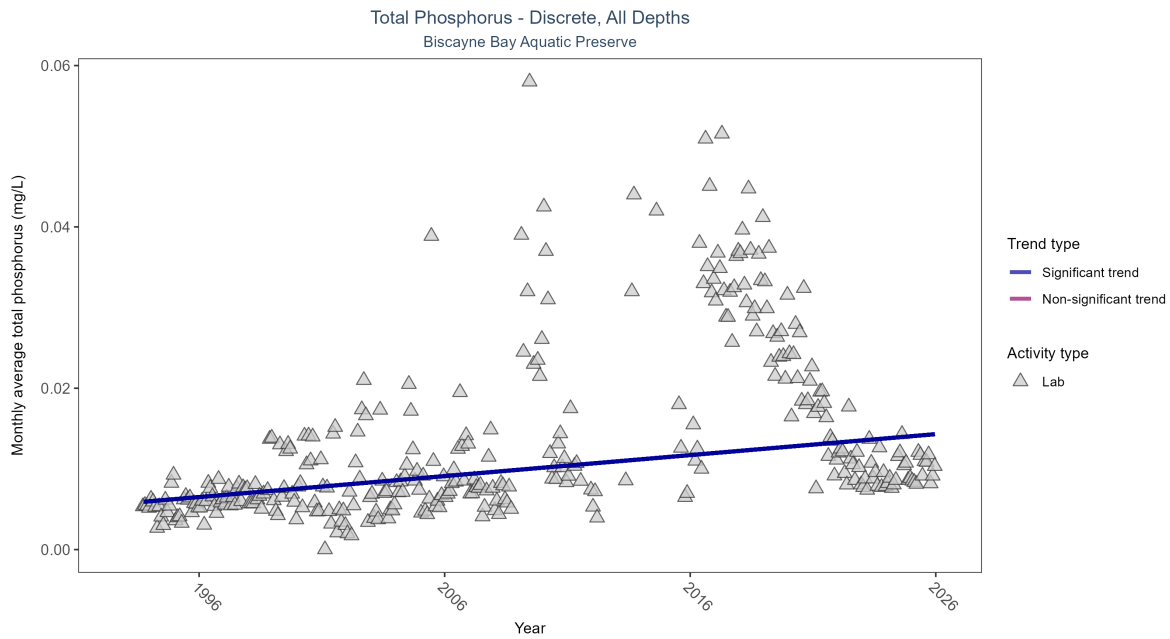


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	9910	33	1993 - 2025	0.009	0.38116	0.00573	0.00026	0

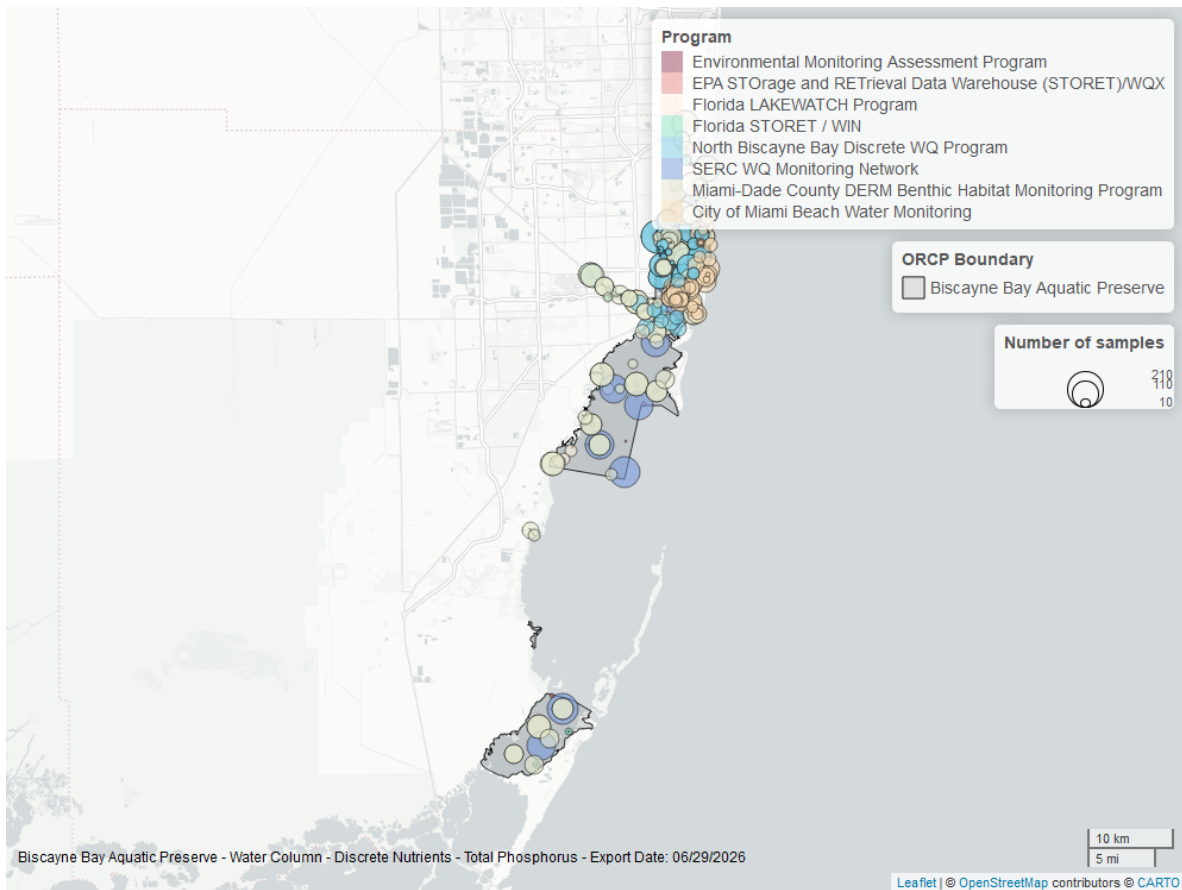


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

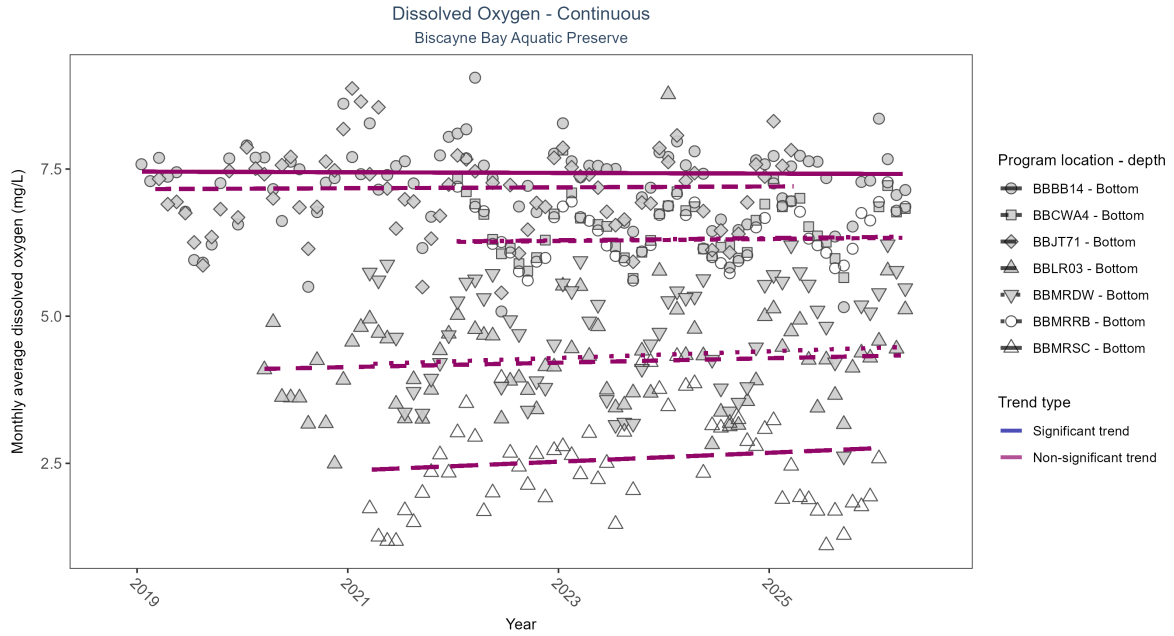


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at seven locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	218187	8	2019 - 2026	7.2	-0.05	7.46	-0.01	0.71
BBCWA4	No detectable trend	126237	5	2022 - 2026	6.5	0.15	6.26	0.02	0.57
BBJT71	No detectable trend	200410	7	2019 - 2025	7.0	0.09	7.16	0.01	0.45
BBLR03	No detectable trend	189960	7	2020 - 2026	4.2	0.09	4.10	0.04	0.40
BBMRDW	No detectable trend	159604	6	2021 - 2026	4.8	0.15	4.18	0.06	0.20
BBMRRB	No detectable trend	134856	5	2022 - 2026	6.4	0.09	6.27	0.02	0.55
BBMRSC	No detectable trend	158733	6	2021 - 2026	2.2	0.08	2.38	0.08	0.52

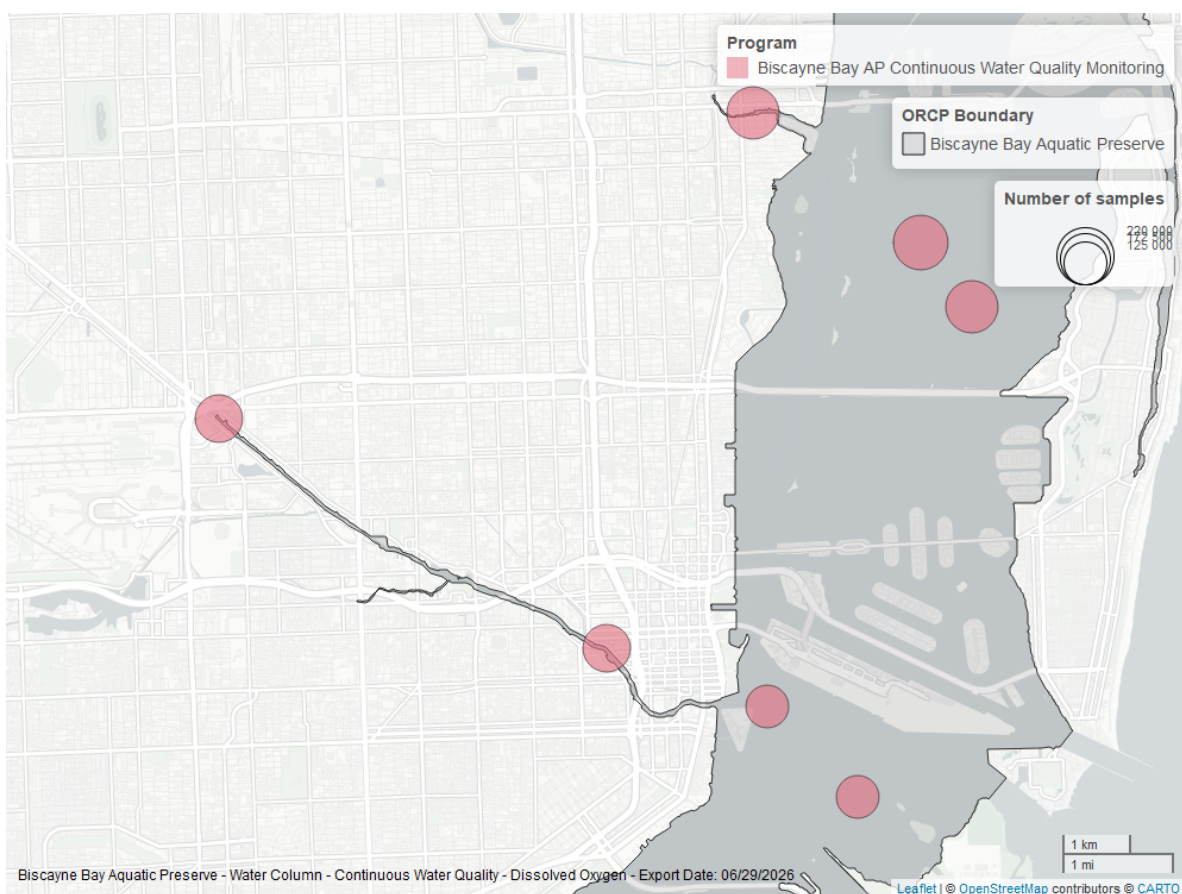


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

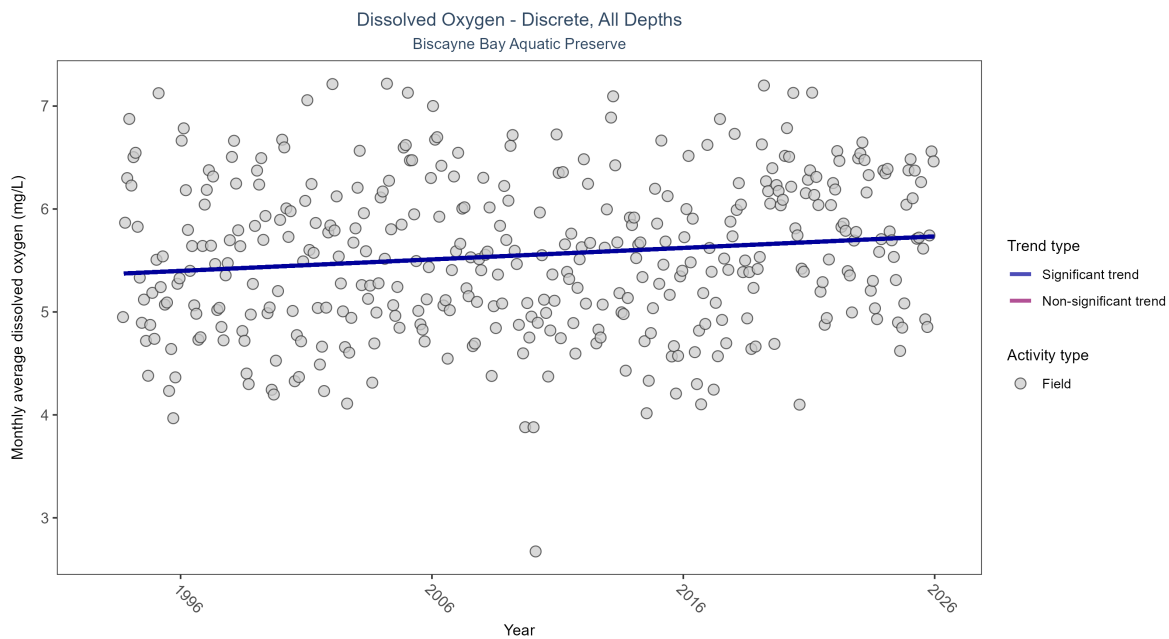


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	45526	33	1993 - 2025	5.81	0.18002	5.36522	0.01116	0

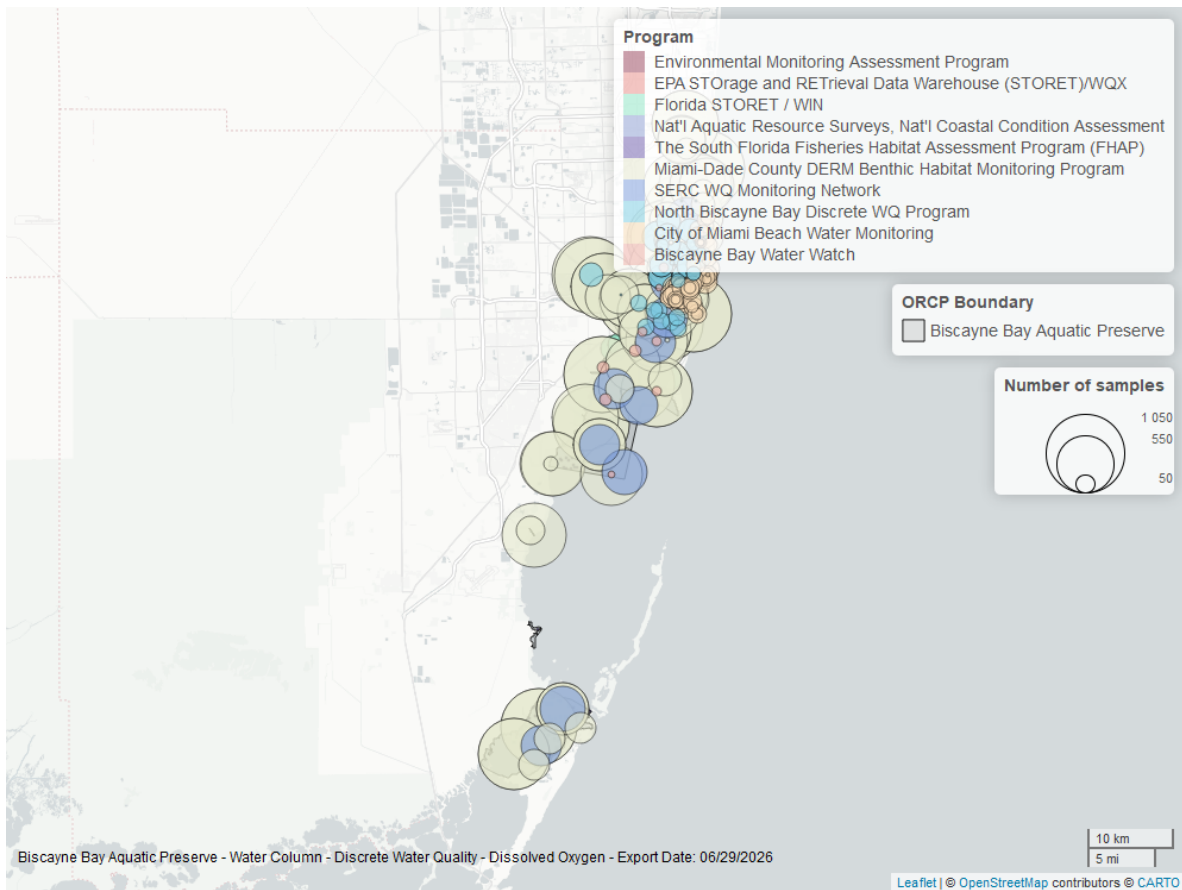


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

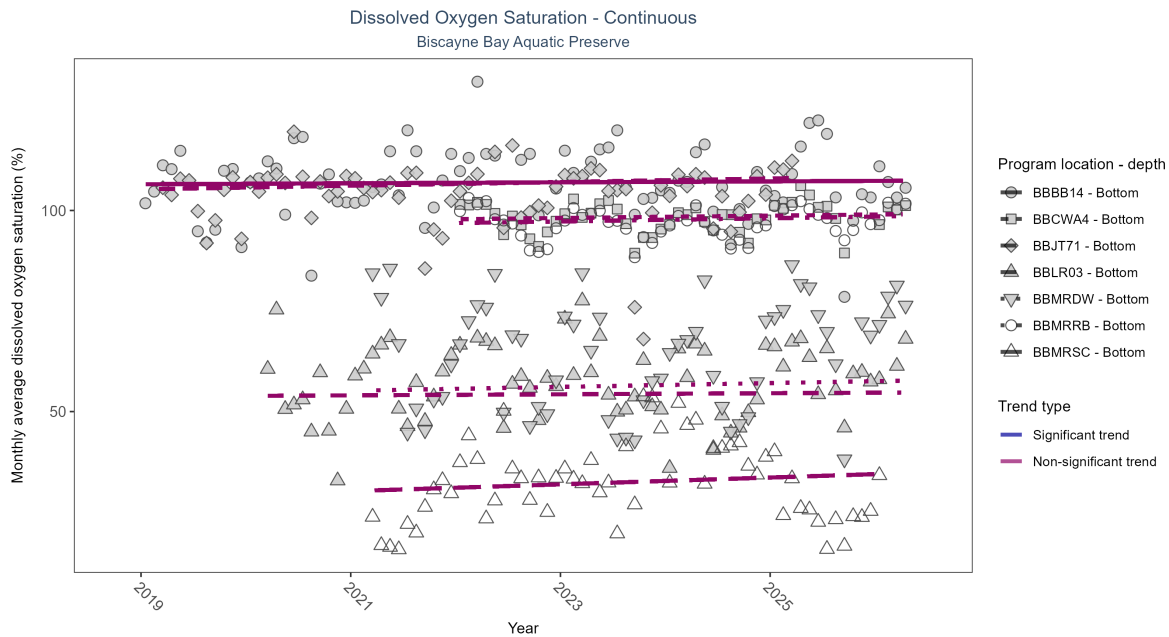


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average dissolved oxygen saturation was observed at seven locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	217943	8	2019 - 2026	104.4	0.01	106.52	0.12	0.90
BBCWA4	No detectable trend	129974	5	2022 - 2026	97.9	0.18	97.84	0.29	0.36
BBJT71	No detectable trend	201561	7	2019 - 2025	101.8	0.14	105.26	0.45	0.20
BBLR03	No detectable trend	189959	7	2020 - 2026	58.0	0.07	53.90	0.13	0.60
BBMRDW	No detectable trend	159605	6	2021 - 2026	63.6	0.07	55.17	0.48	0.54
BBMRRB	No detectable trend	134864	5	2022 - 2026	96.3	0.22	96.87	0.40	0.20
BBMRSC	No detectable trend	158630	6	2021 - 2026	28.9	0.08	30.21	0.85	0.52

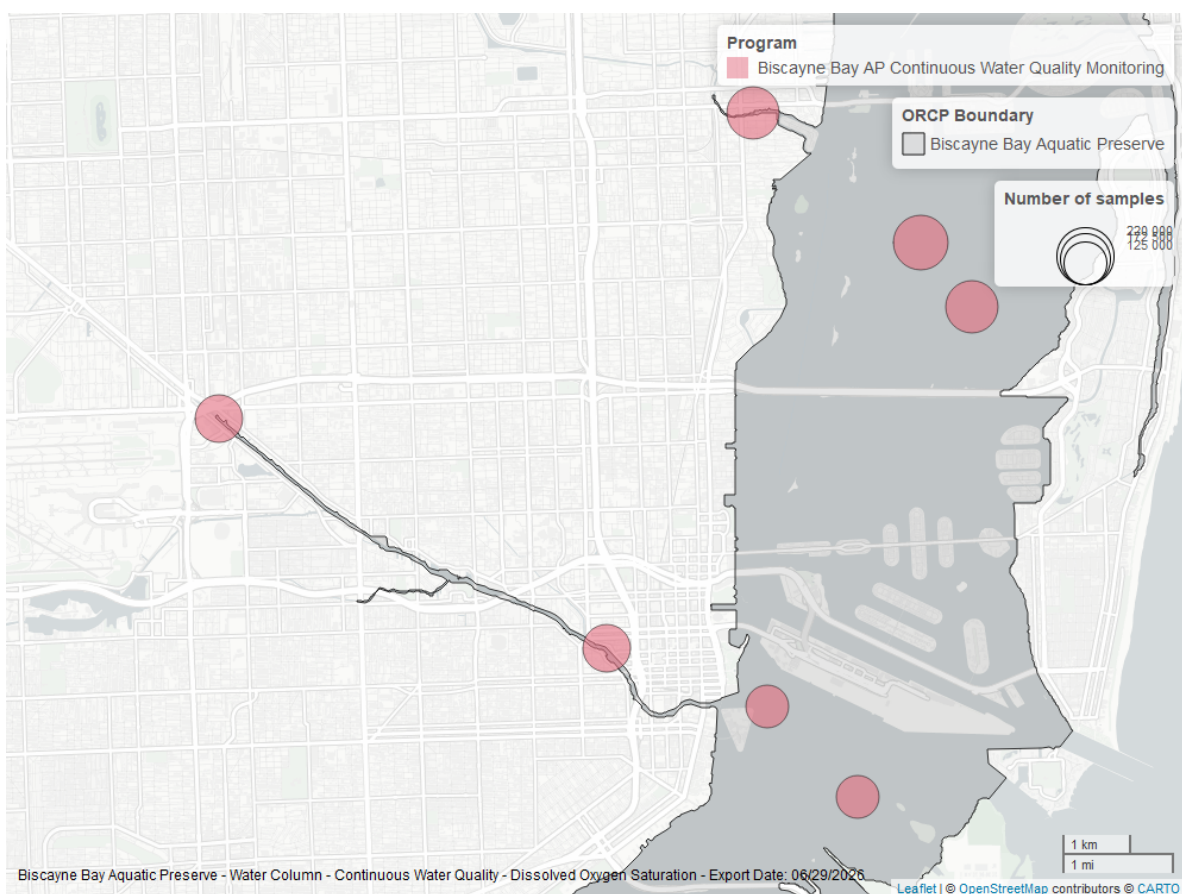


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

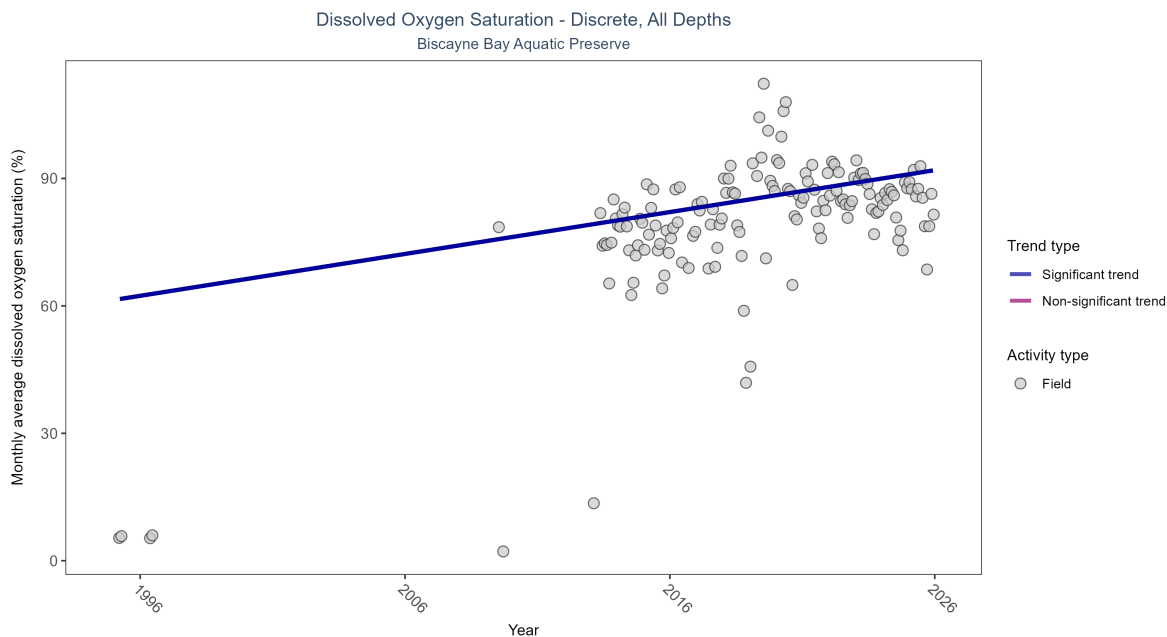


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.99% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	15094	16	1995 - 2025	88.7	0.38467	61.38306	0.9861	0

\end{table}

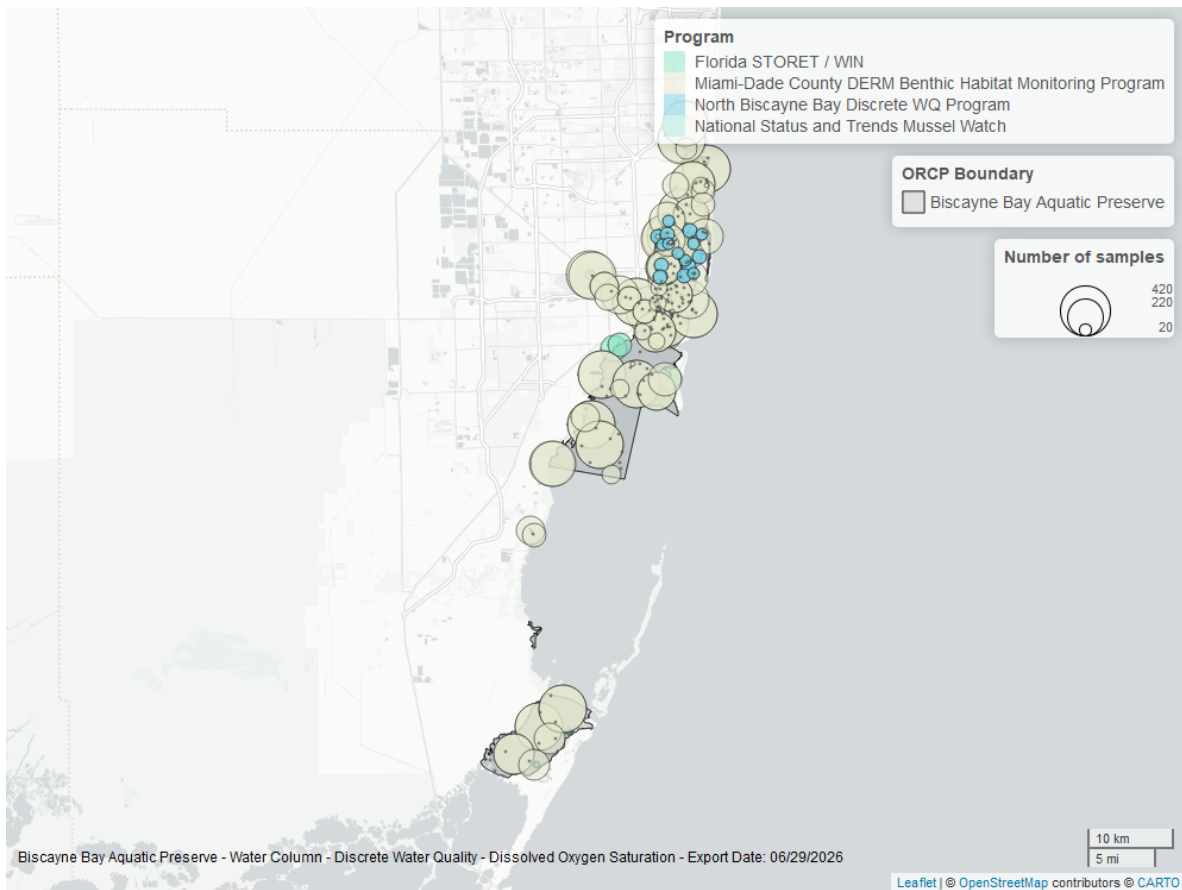


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

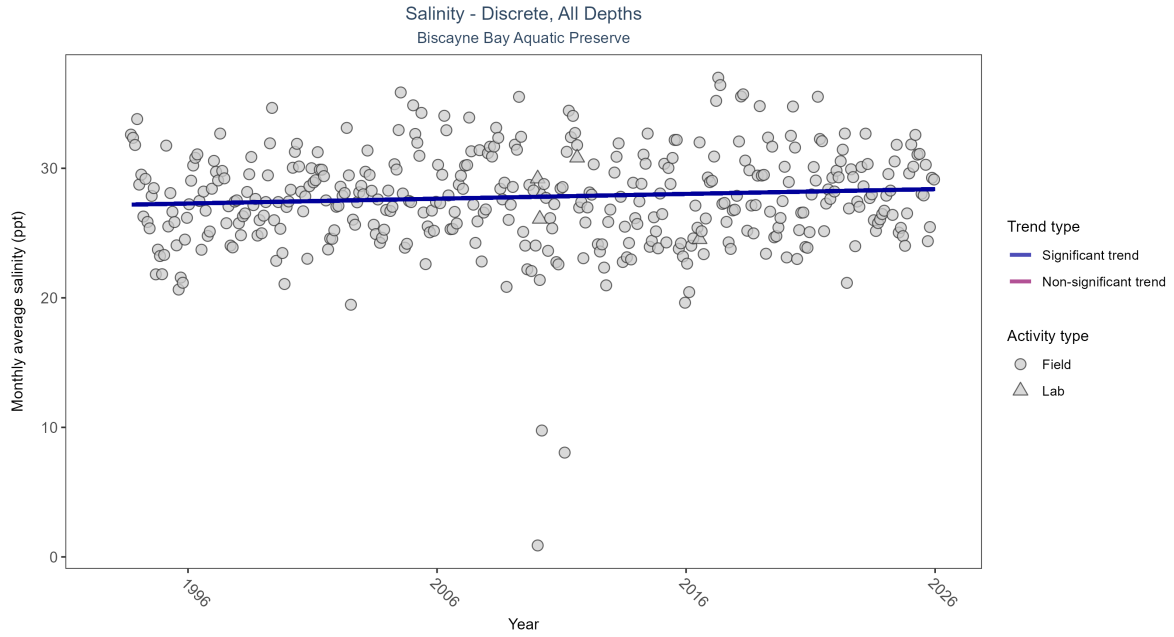


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Monthly average salinity increased by 0.04 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	50494	33	1993 - 2025	31.4	0.0877	27.16942	0.03705	0.0144

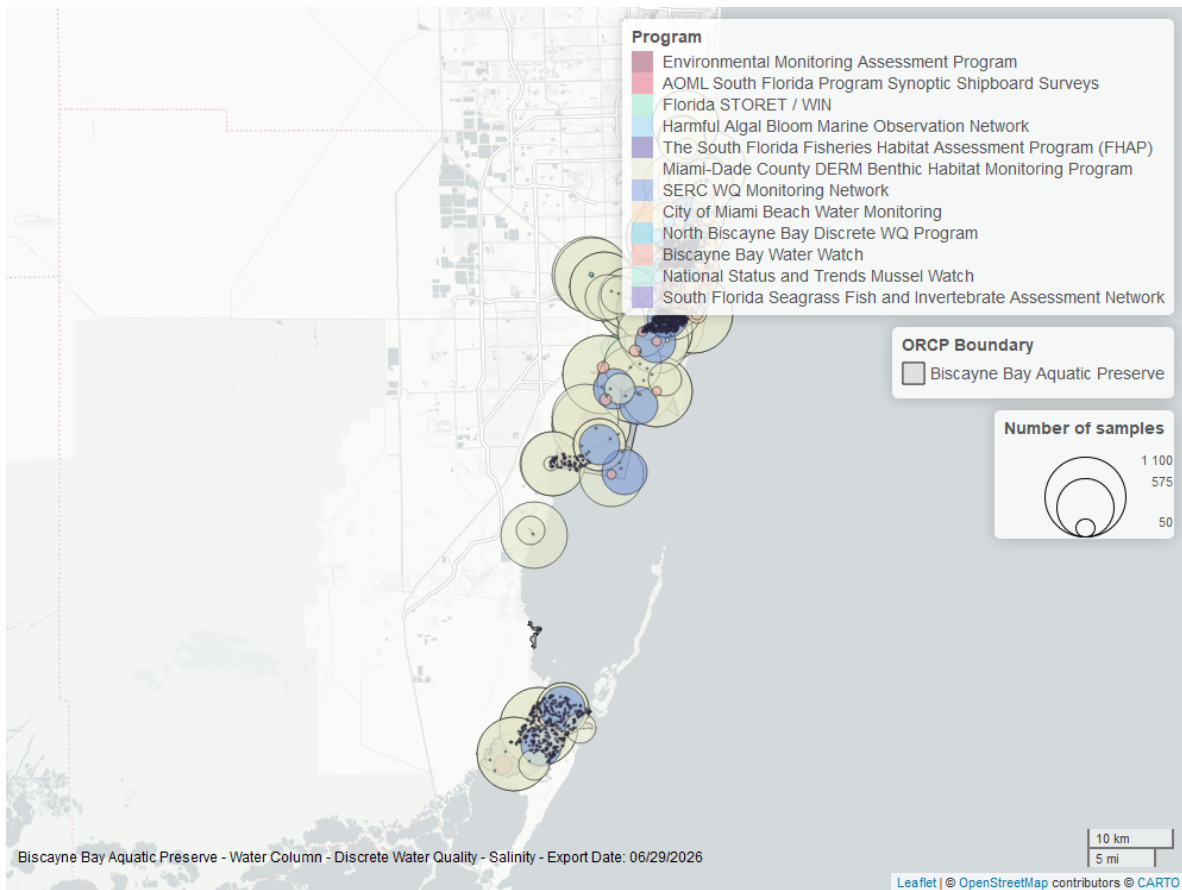


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

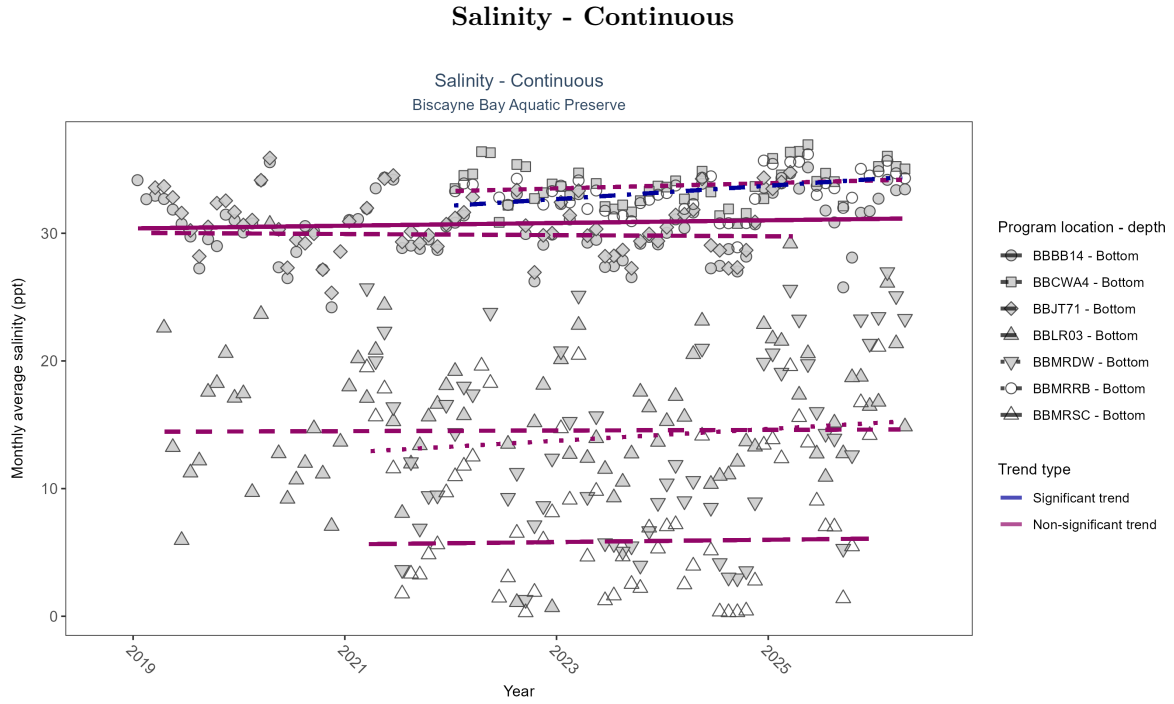


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At one program location, monthly average salinity increased by 0.52 ppt per year. No detectable change in monthly average salinity was observed at six locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	204077	8	2019 - 2026	30.7	0.08	30.38	0.11	0.40
BBCWA4	No detectable trend	112816	5	2022 - 2026	33.8	0.16	33.30	0.21	0.13
BBJT71	No detectable trend	178691	7	2019 - 2025	30.8	-0.02	30.03	-0.04	0.91
BBLR03	No detectable trend	195042	8	2019 - 2026	19.3	0.01	14.45	0.02	0.96
BBMRDW	No detectable trend	151607	6	2021 - 2026	13.0	0.10	12.82	0.46	0.48
BBMRRB	Increasing trend	118477	5	2022 - 2026	33.5	0.42	32.16	0.52	0.00
BBMRSC	No detectable trend	150427	6	2021 - 2026	5.6	0.03	5.63	0.09	0.83

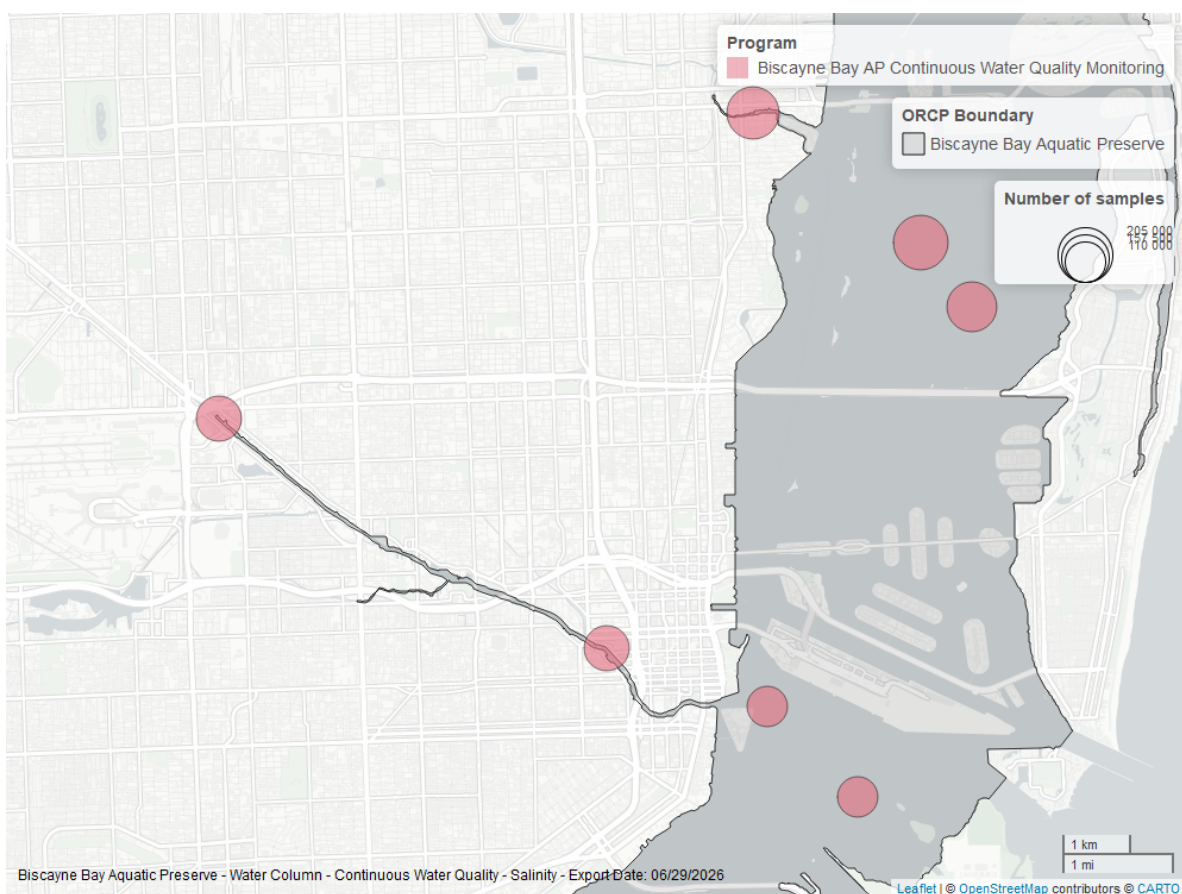


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

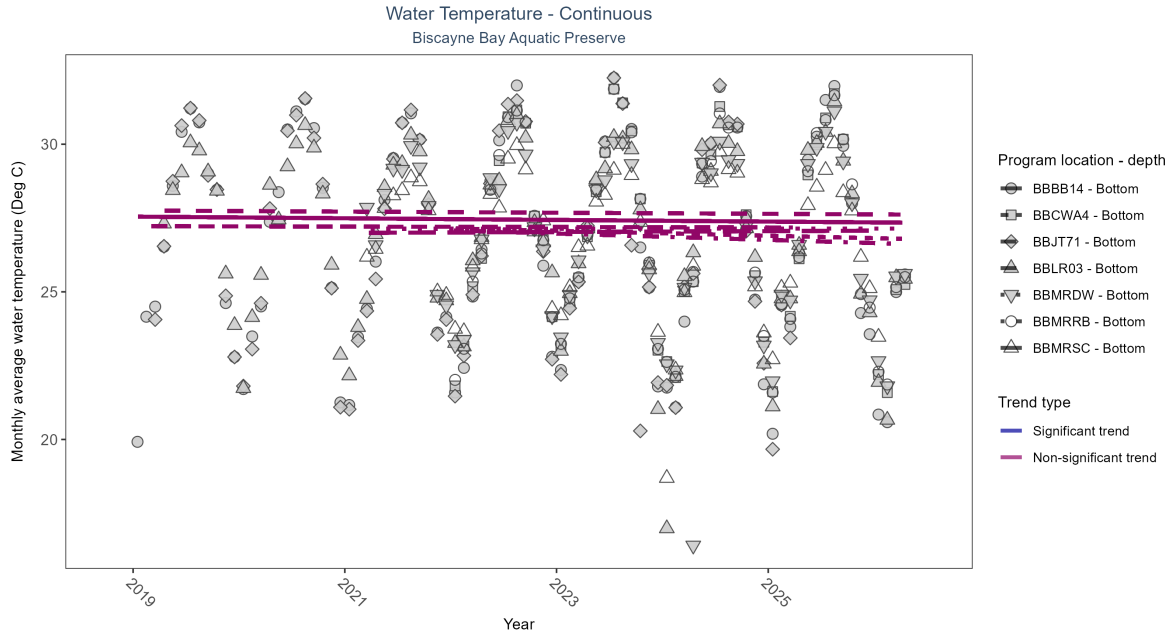


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: No detectable change in monthly average water temperature was observed at seven locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBB14	No detectable trend	226327	8	2019 - 2026	26.8	-0.03	27.55	-0.03	0.65
BBCWA4	No detectable trend	130160	5	2022 - 2026	27.0	-0.18	27.15	-0.08	0.20
BBJT71	No detectable trend	208542	7	2019 - 2025	27.1	-0.03	27.22	-0.01	0.87
BBLR03	No detectable trend	222607	8	2019 - 2026	27.6	-0.02	27.76	-0.02	0.73
BBMRDW	No detectable trend	167391	6	2021 - 2026	27.2	0.02	27.14	0.00	1.00
BBMRRB	No detectable trend	139606	5	2022 - 2026	26.8	-0.13	27.08	-0.11	0.26
BBMRSC	No detectable trend	163580	6	2021 - 2026	27.2	0.03	27.00	0.02	0.83

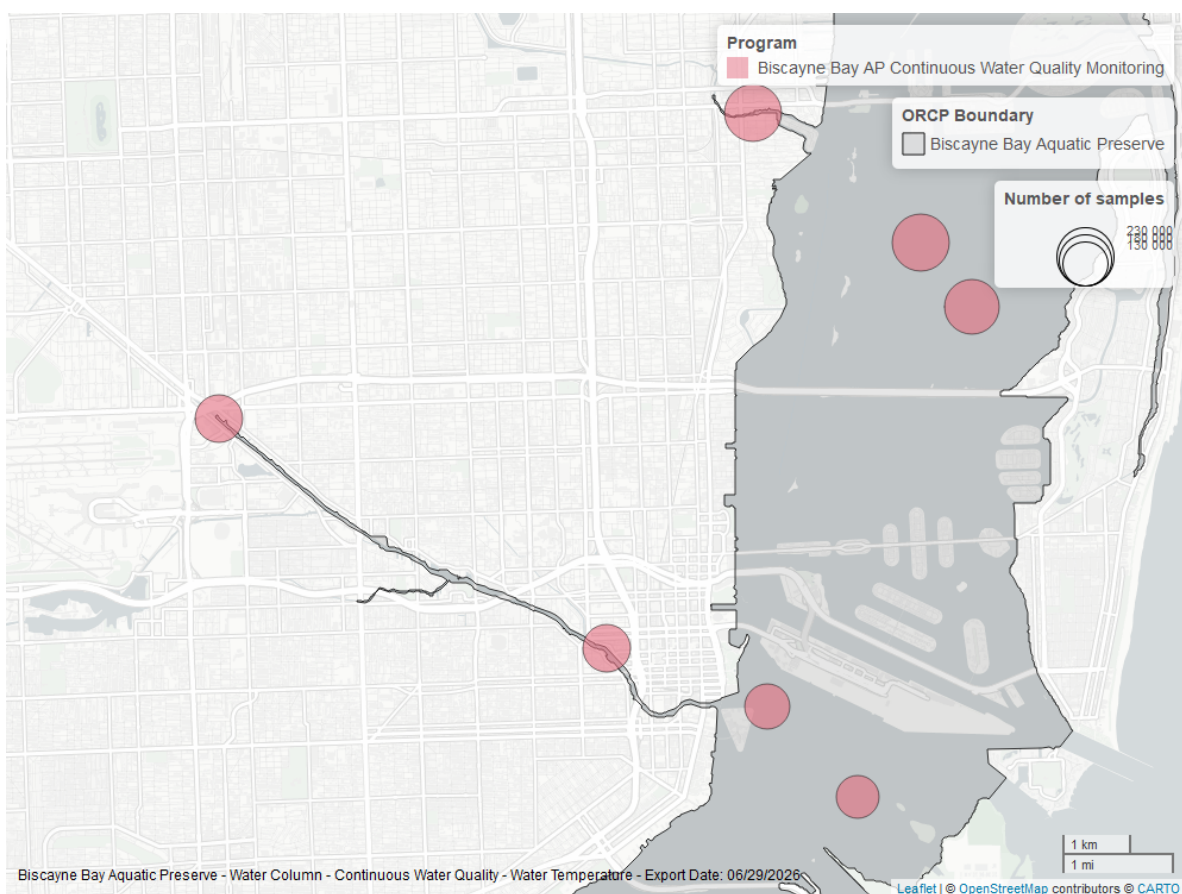


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

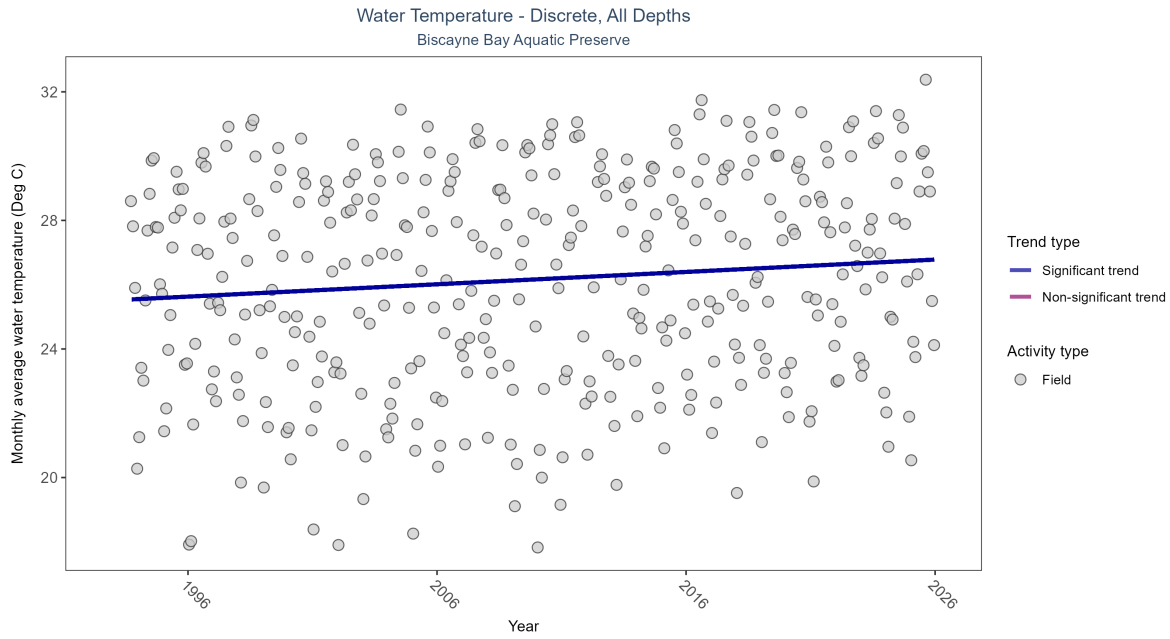


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Monthly average water temperature increased by 0.04Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	50776	33	1993 - 2025	26.63	0.22669	25.51344	0.0384	0

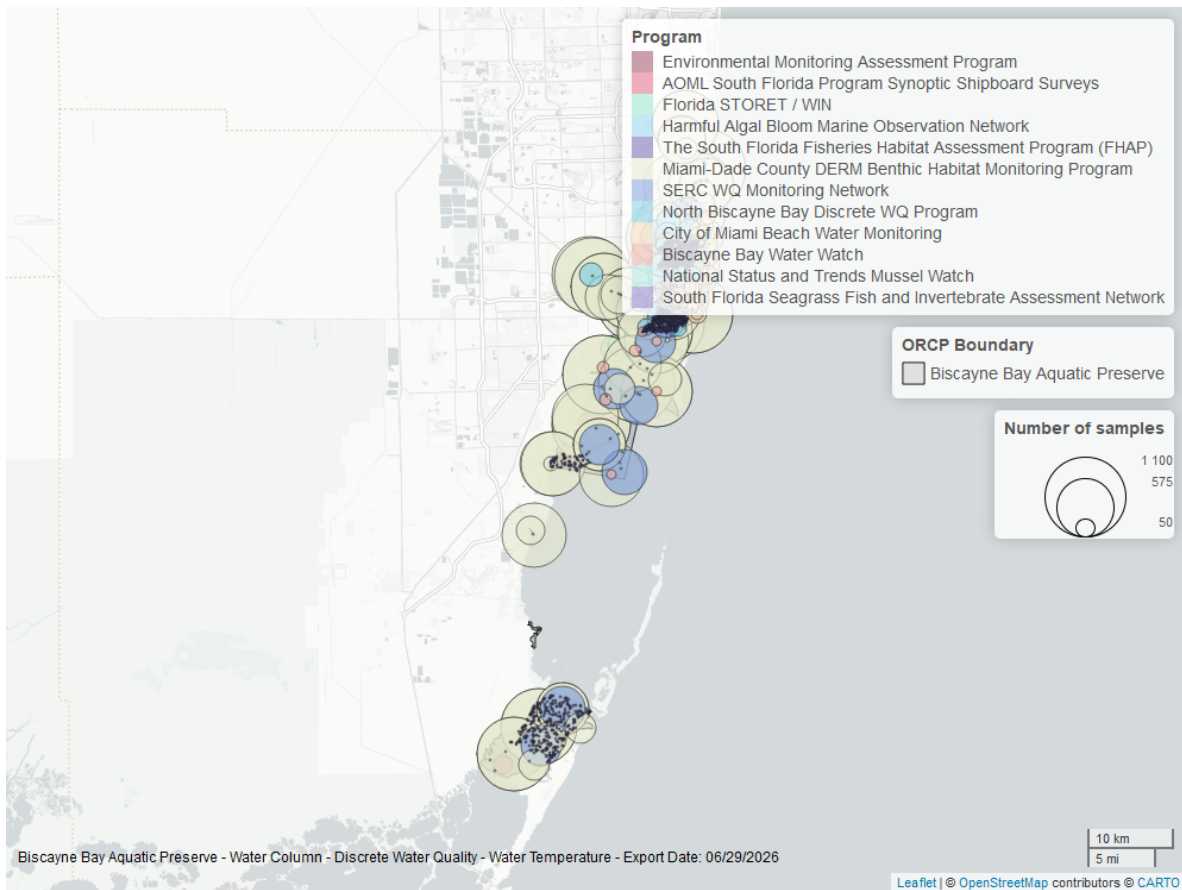


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

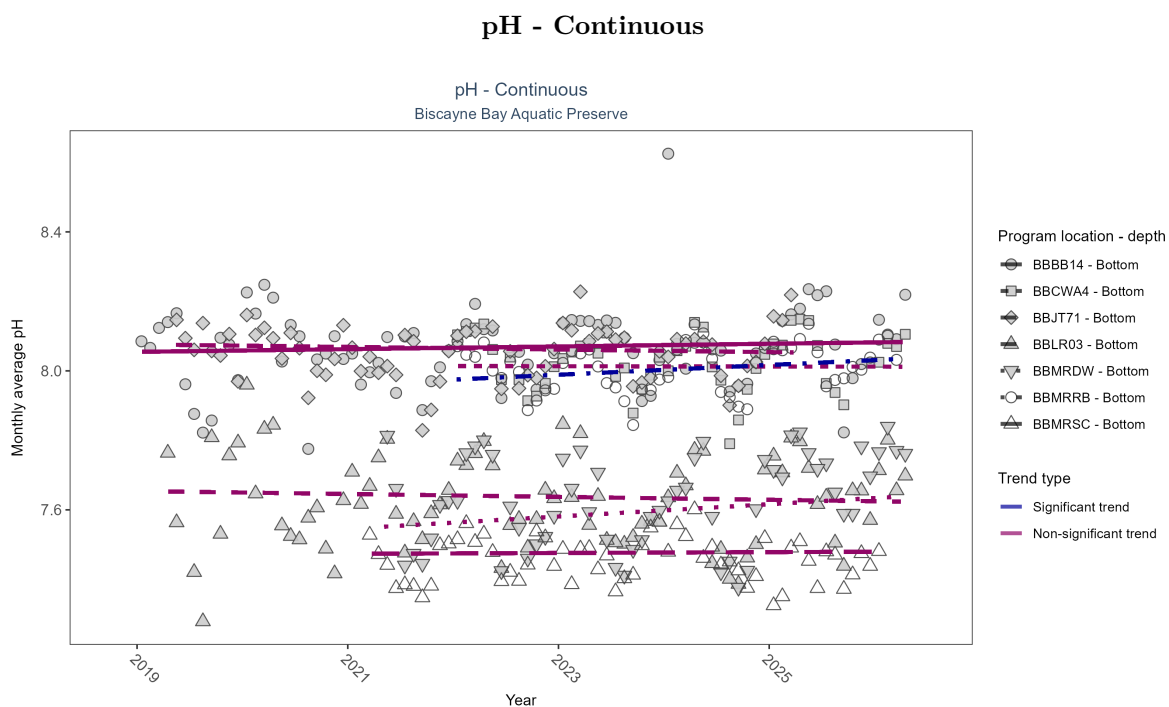


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At one program location, monthly average pH increased by 0.01 pH units per year. No detectable change in monthly average pH was observed at six locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	219134	8	2019 - 2026	8.1	0.08	8.05	0.00	0.39
BBCWA4	No detectable trend	130007	5	2022 - 2026	8.0	0.05	8.01	0.00	1.00
BBJT71	No detectable trend	190835	7	2019 - 2025	8.1	-0.09	8.08	0.00	0.43
BBLR03	No detectable trend	211851	8	2019 - 2026	7.6	-0.06	7.65	0.00	0.49
BBMRDW	No detectable trend	158122	6	2021 - 2026	7.6	0.18	7.55	0.02	0.14
BBMRRB	Increasing trend	132608	5	2022 - 2026	8.0	0.43	7.97	0.01	0.00
BBMRSC	No detectable trend	157812	6	2021 - 2026	7.4	0.04	7.47	0.00	0.72

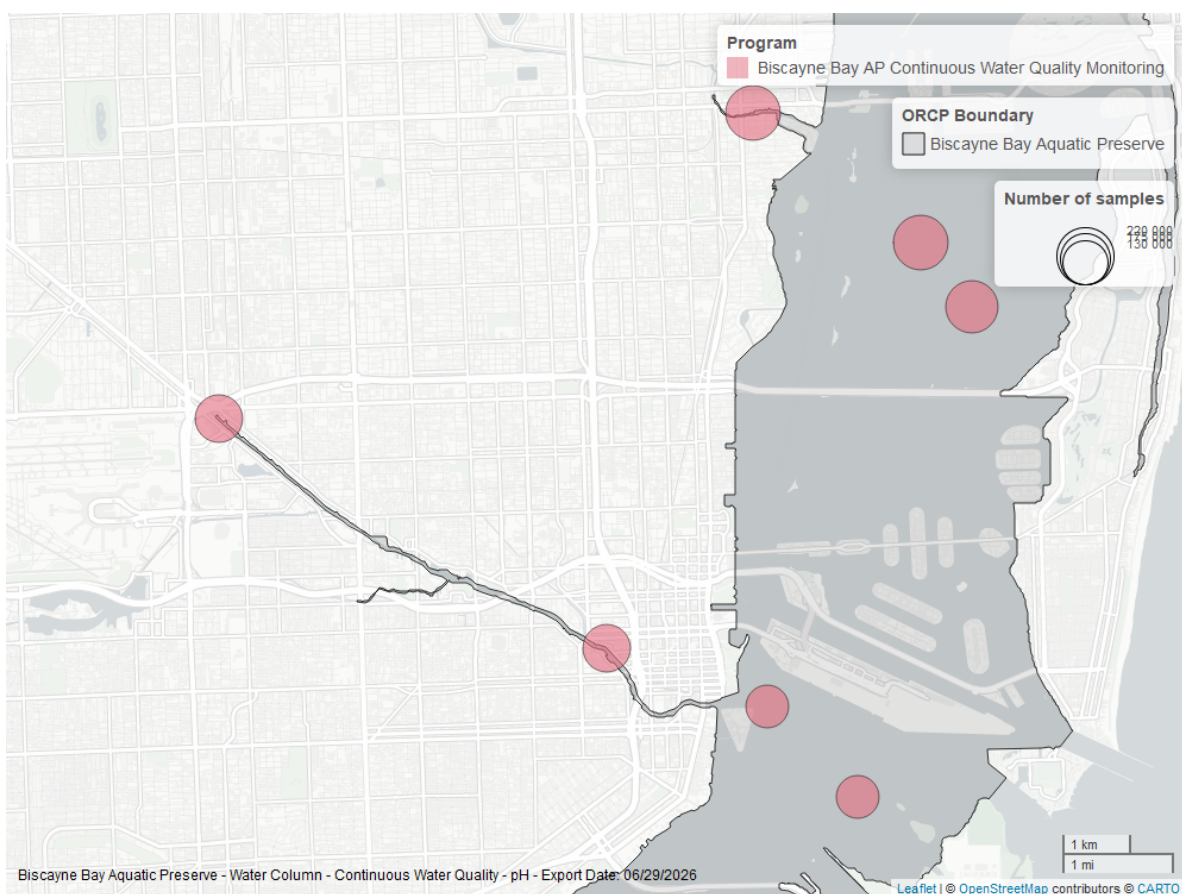


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

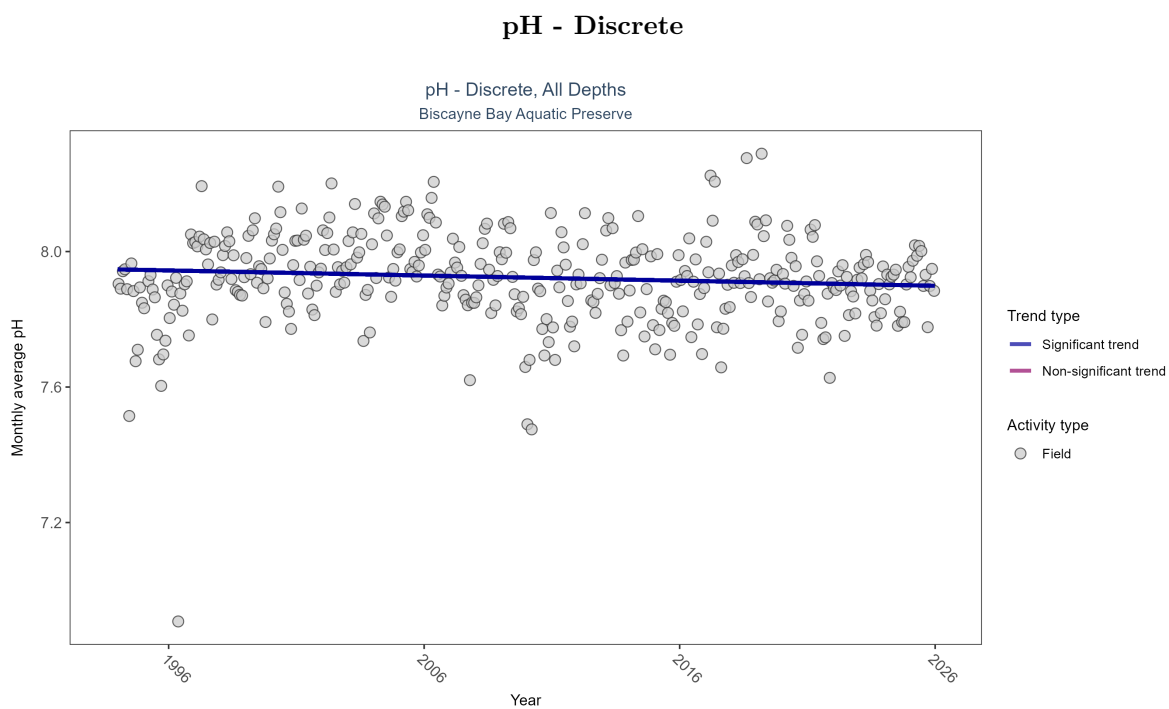


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	43921	32	1994 - 2025	7.98	-0.10471	7.94722	-0.00151	0.0037

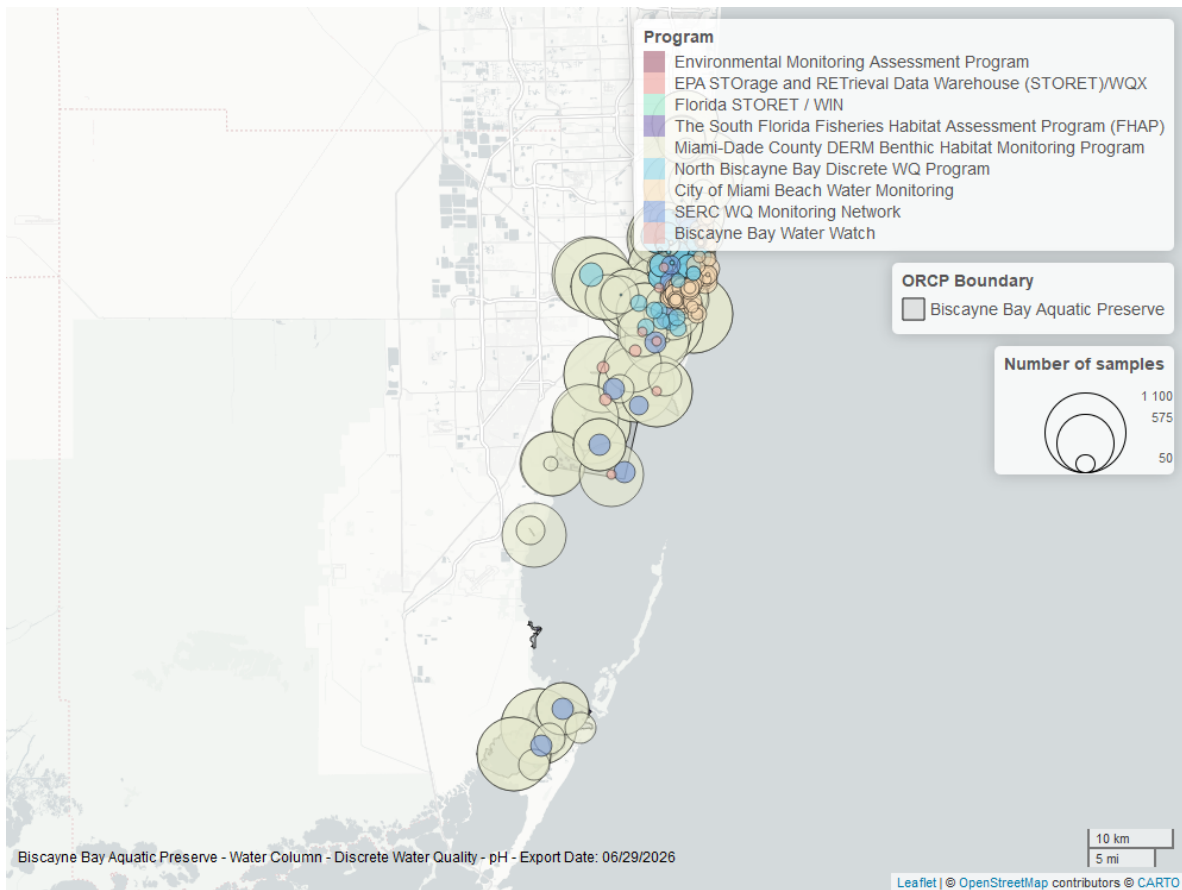


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

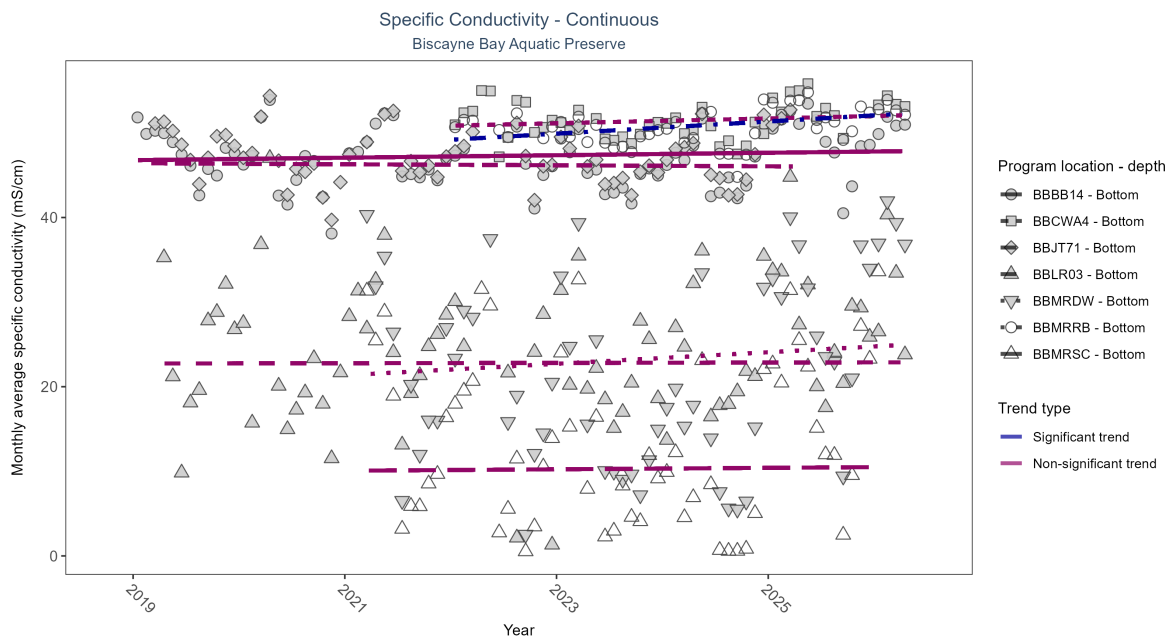


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At one program location, monthly average specific conductivity increased by 0.72 mS/cm per year. No detectable change in monthly average specific conductivity was observed at six locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	204078	8	2019 - 2026	47.25	0.06	46.78	0.15	0.51
BBCWA4	No detectable trend	112835	5	2022 - 2026	51.48	0.13	50.85	0.29	0.19
BBJT71	No detectable trend	178691	7	2019 - 2025	47.42	-0.02	46.41	-0.06	0.91
BBLR03	No detectable trend	196867	8	2019 - 2026	30.36	0.00	22.74	0.02	1.00
BBMRDW	No detectable trend	151609	6	2021 - 2026	21.71	0.11	21.35	0.68	0.40
BBMRRB	Increasing trend	118478	5	2022 - 2026	51.07	0.42	49.19	0.72	0.00
BBMRSC	No detectable trend	151200	6	2021 - 2026	9.89	0.05	10.07	0.09	0.72

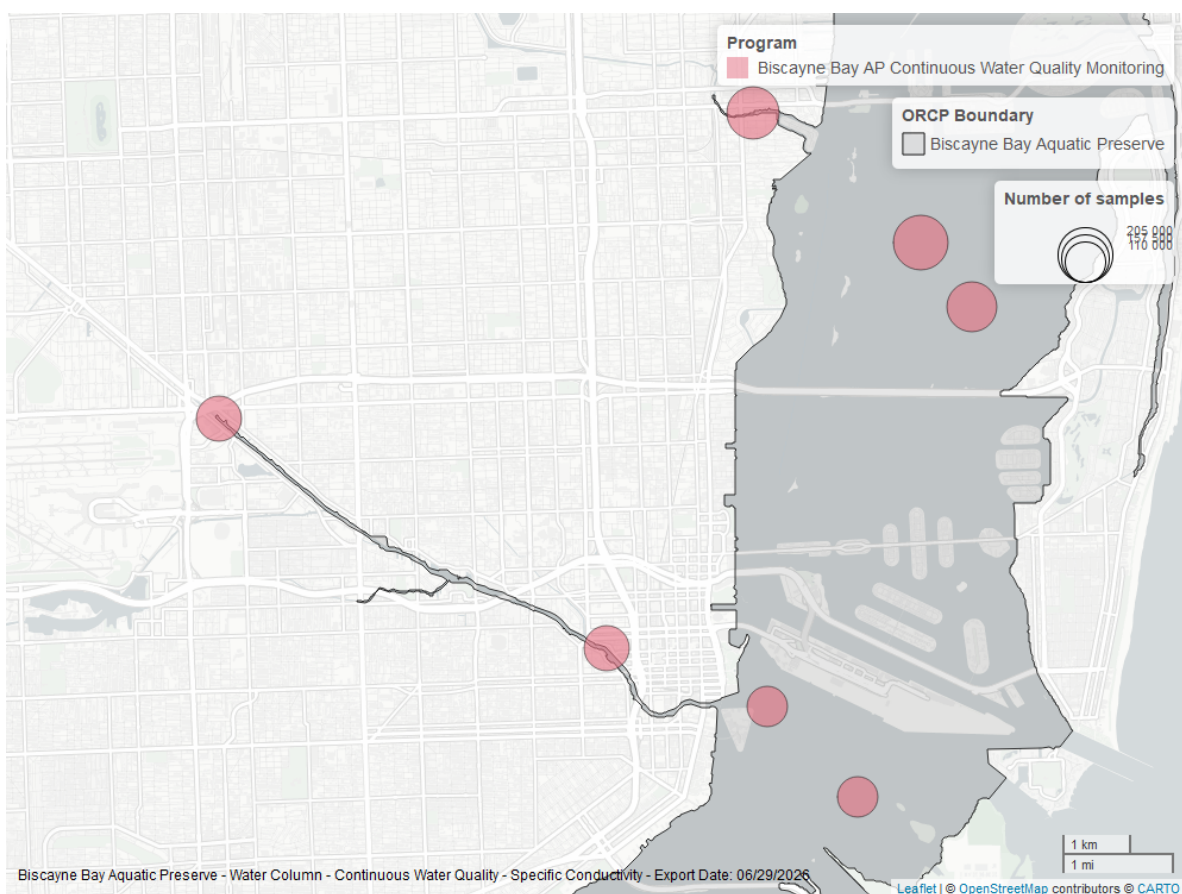


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

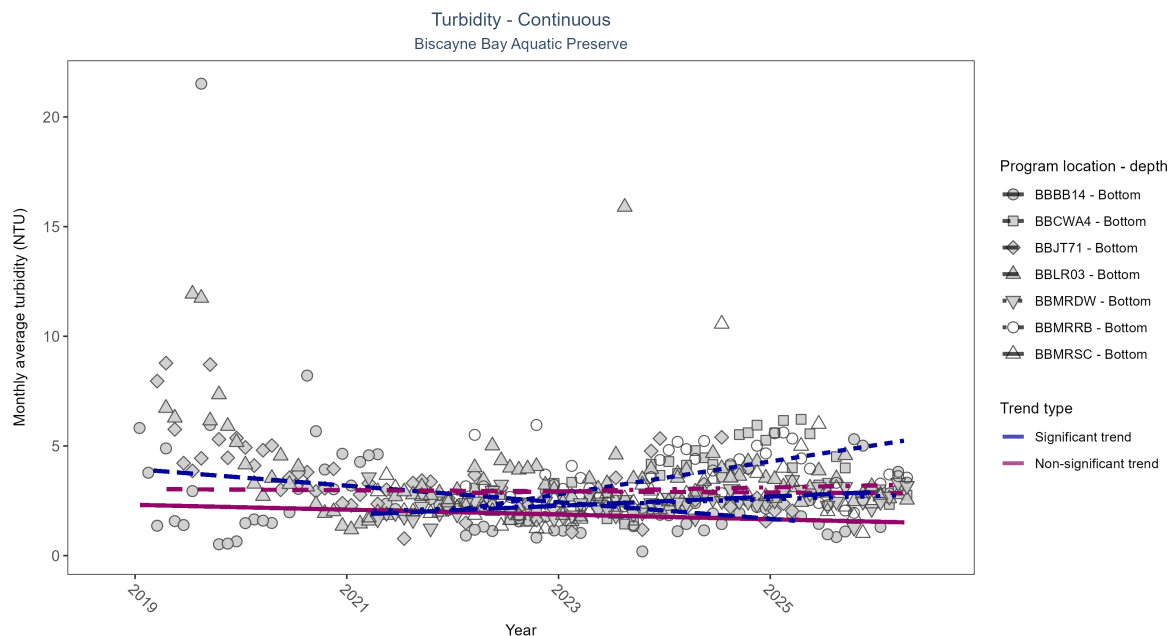


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At three program locations, monthly average turbidity increased between 0.14 and 0.75 NTU per year. At one program location, monthly average turbidity decreased by 0.38 NTU per year. No detectable change in monthly average turbidity was observed at three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No detectable trend	205753	8	2019 - 2026	2	-0.13	2.31	-0.11	0.17
BBCWA4	Increasing trend	119922	5	2022 - 2026	3	0.52	2.04	0.75	0.00
BBJT71	Decreasing trend	195318	7	2019 - 2025	3	-0.46	3.94	-0.38	0.00
BBLR03	No detectable trend	212624	8	2019 - 2026	3	-0.04	3.04	-0.03	0.60
BBMRDW	Increasing trend	147906	6	2021 - 2026	2	0.36	2.00	0.14	0.01
BBMRRB	No detectable trend	132581	5	2022 - 2026	3	0.09	2.84	0.09	0.44
BBMRSC	Increasing trend	141977	6	2021 - 2026	2	0.30	1.85	0.21	0.01

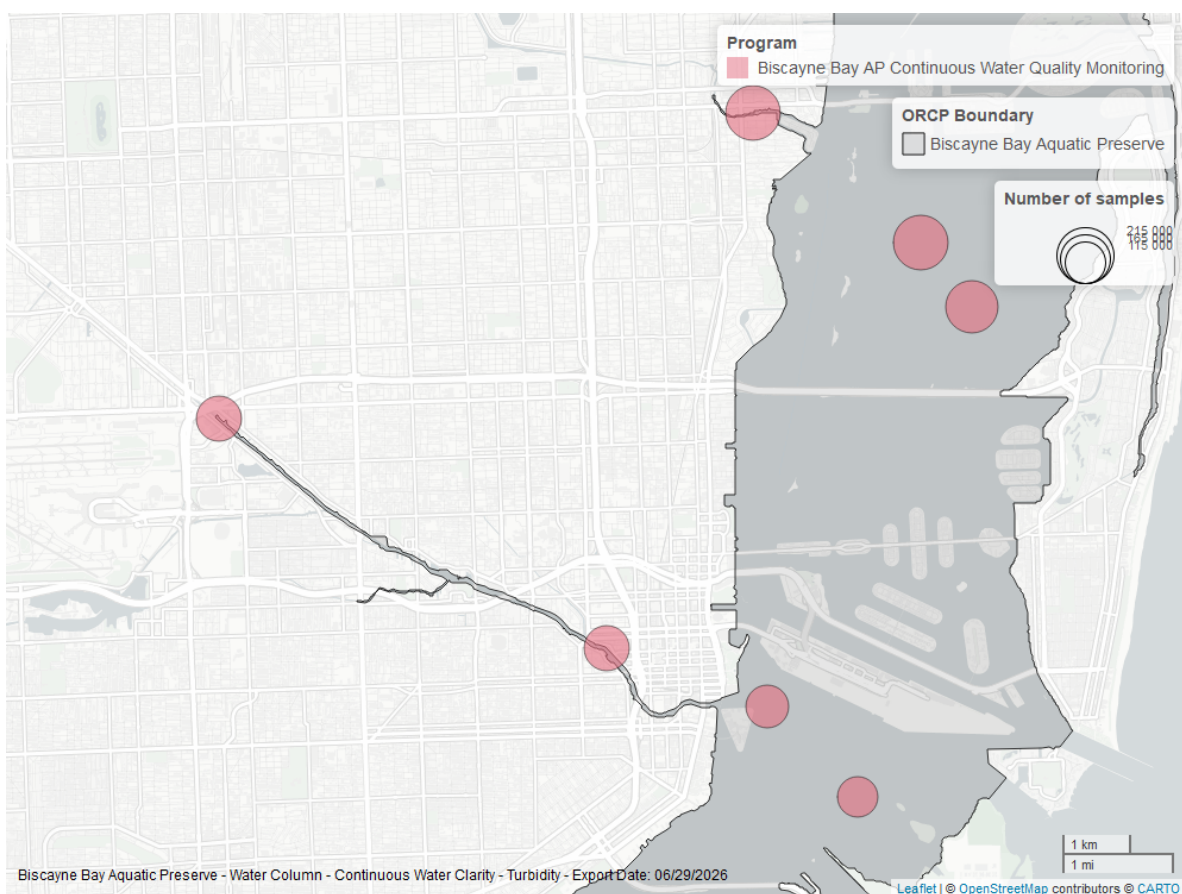


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

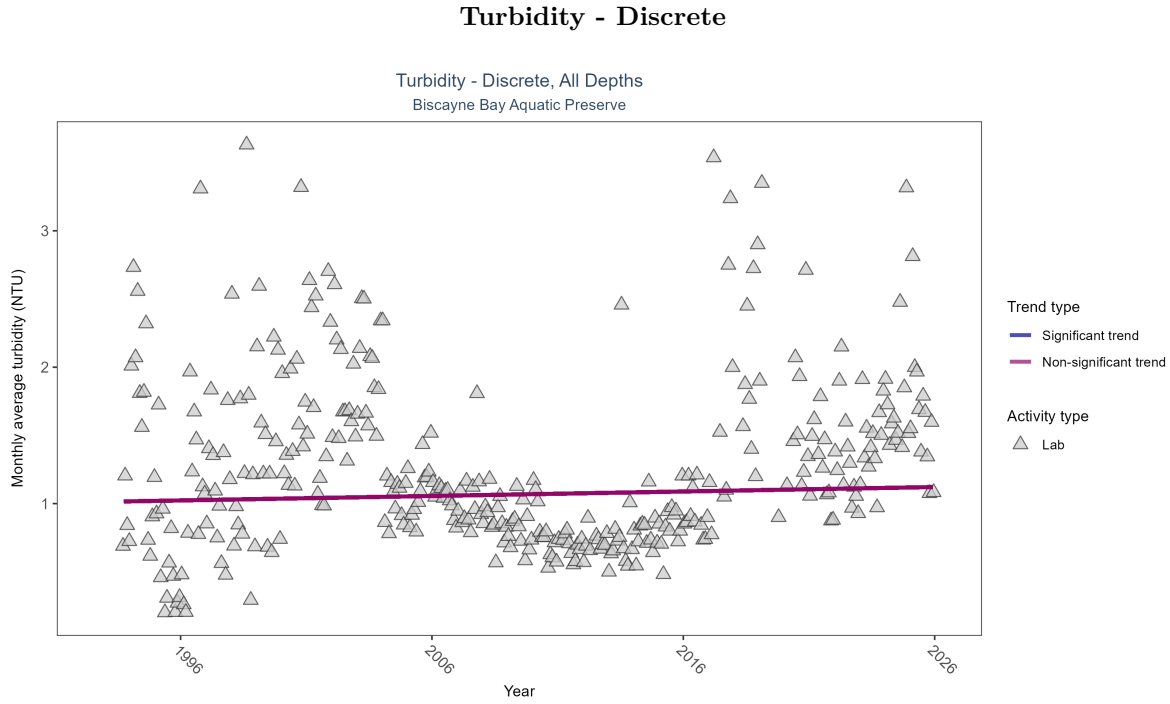


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Turbidity showed no detectable trend between 1993 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	13954	33	1993 - 2025	0.9	0.03996	1.01324	0.00328	0.2562

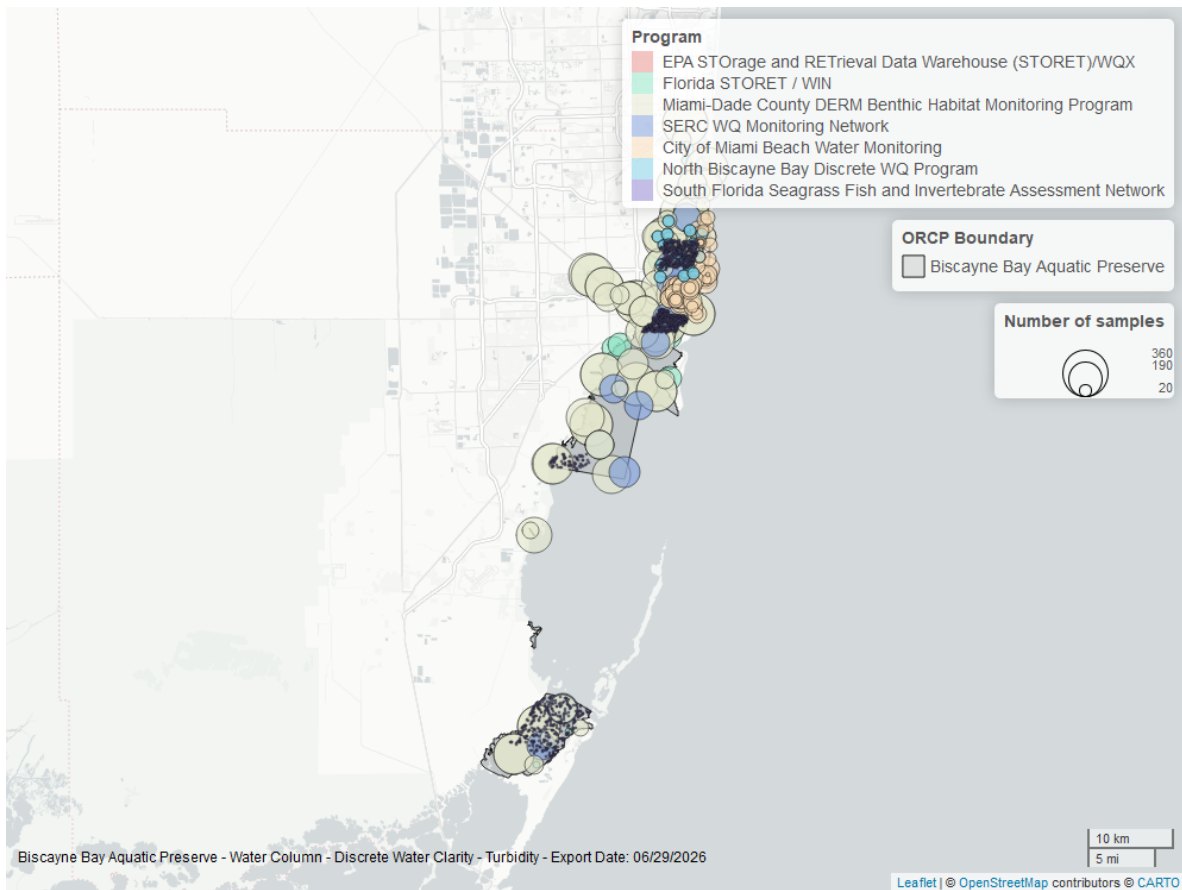


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

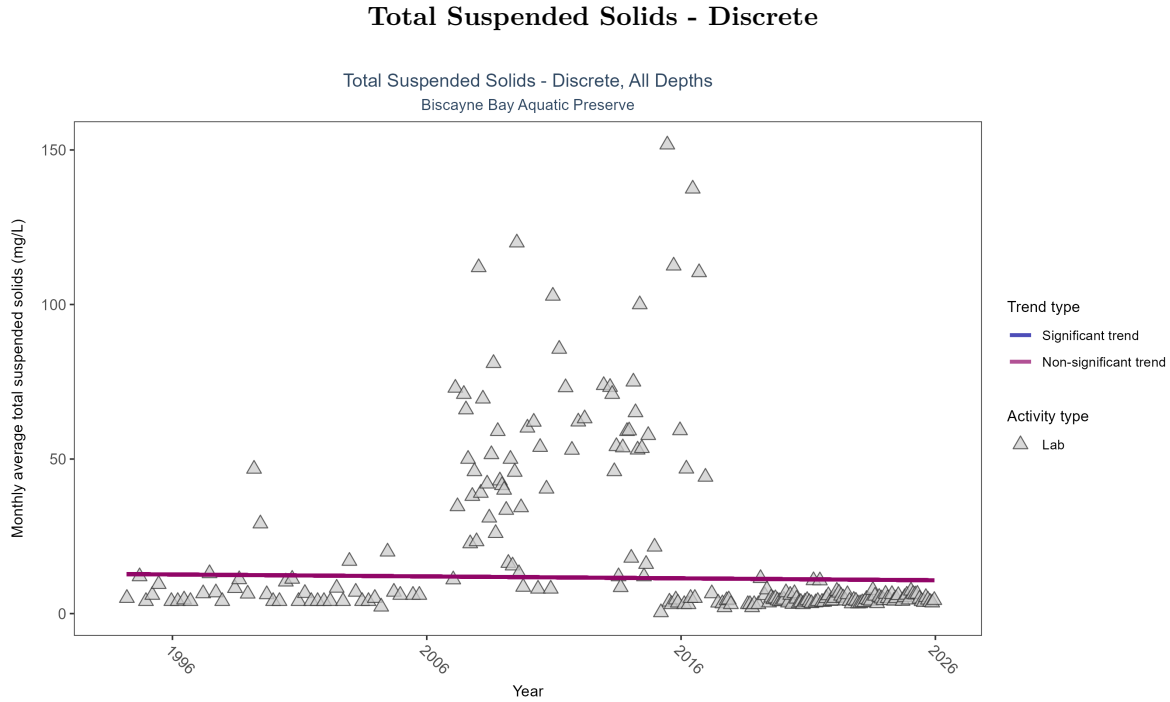


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: Total suspended solids showed no detectable trend between 1994 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2951	31	1994 - 2025	5	-0.15505	12.77674	-0.0625	0.0996

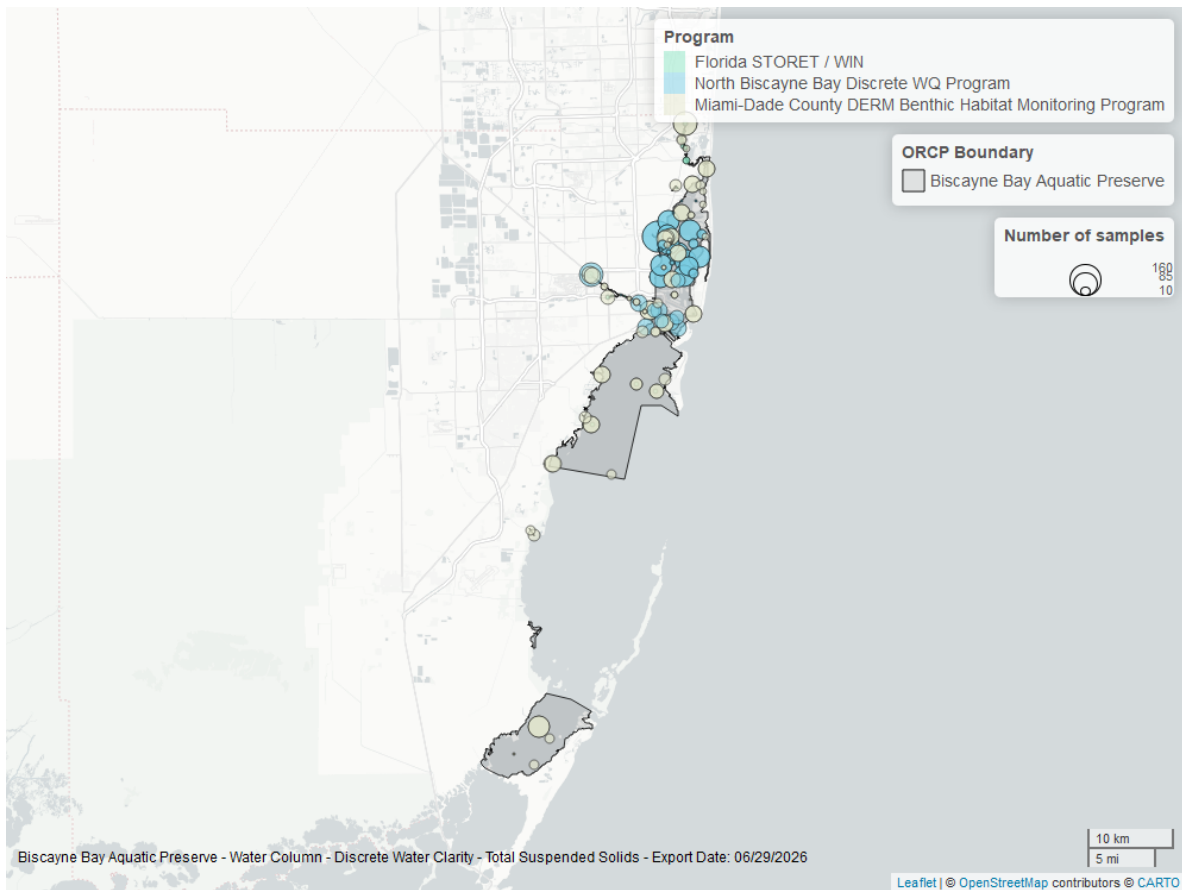


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

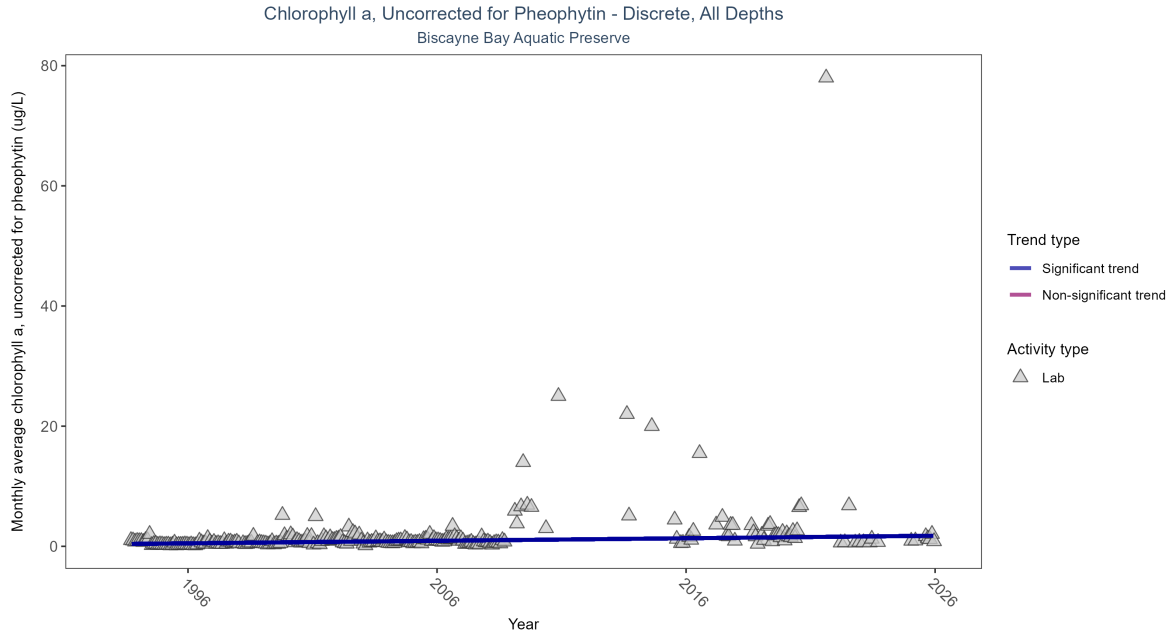


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.04 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2513	30	1993 - 2025	0.70354	0.37519	0.37401	0.04207	0

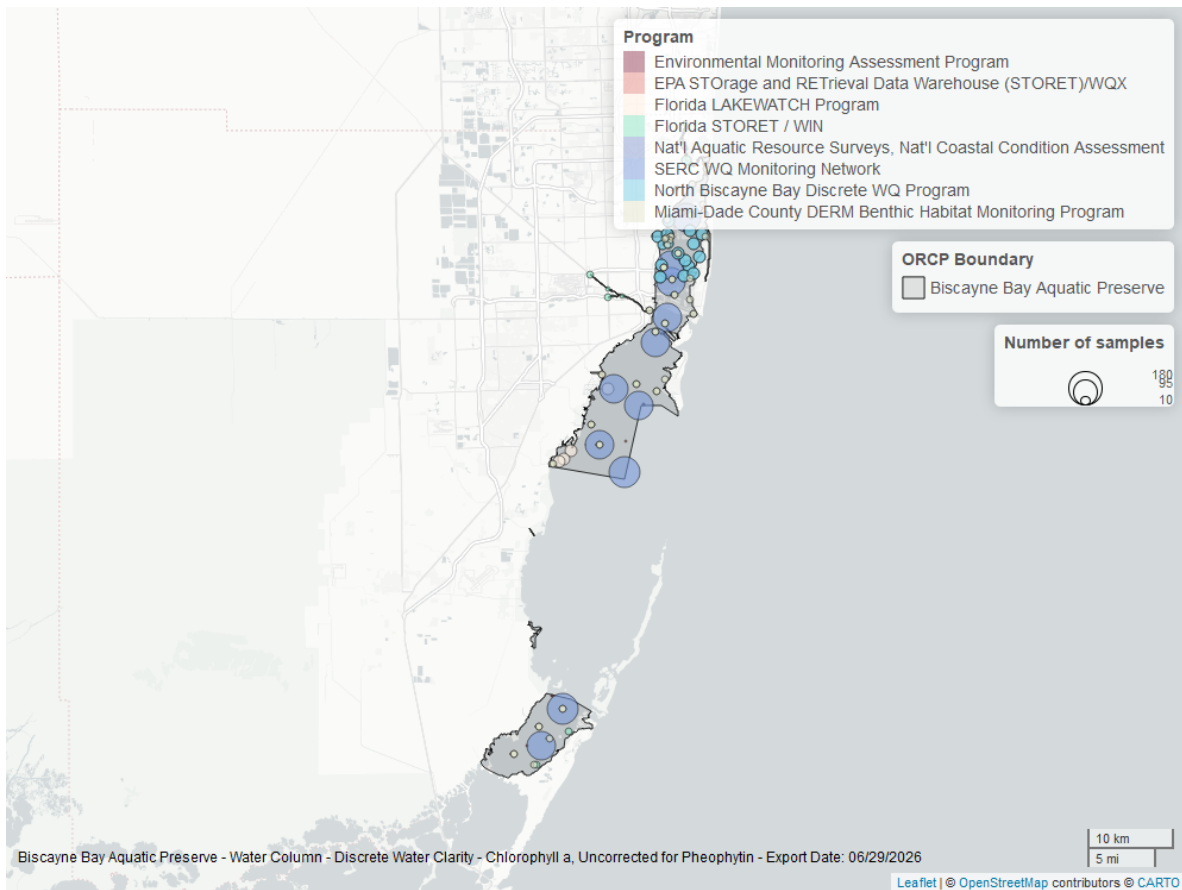


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

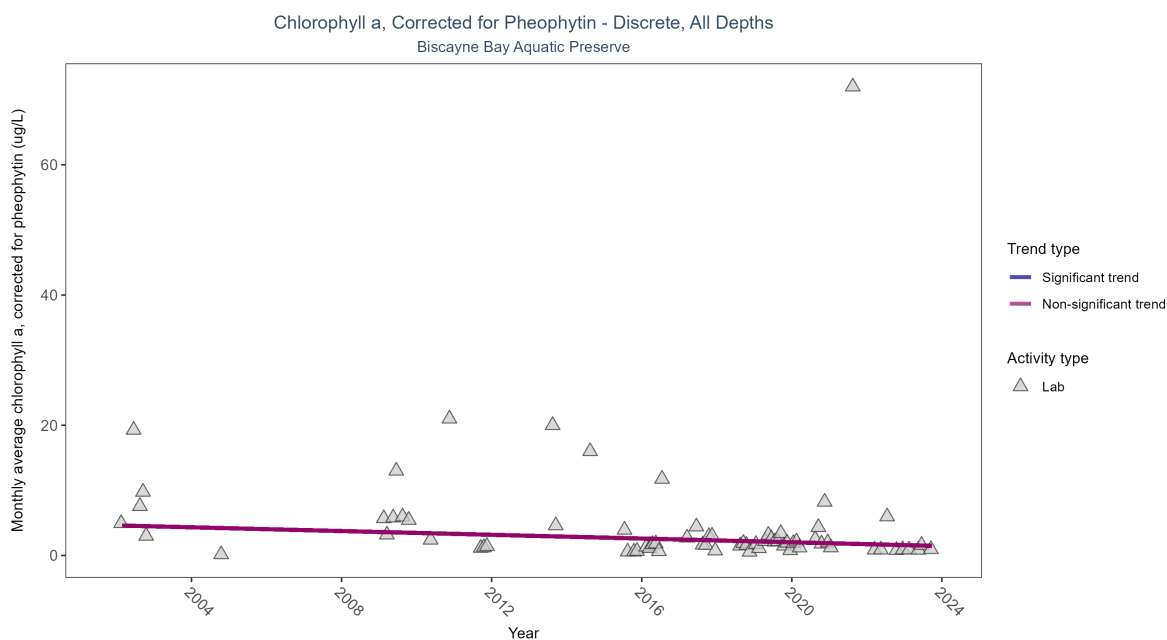


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2023.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	492	16	2002 - 2023	1.4	-0.20161	4.61564	-0.14393	0.0505

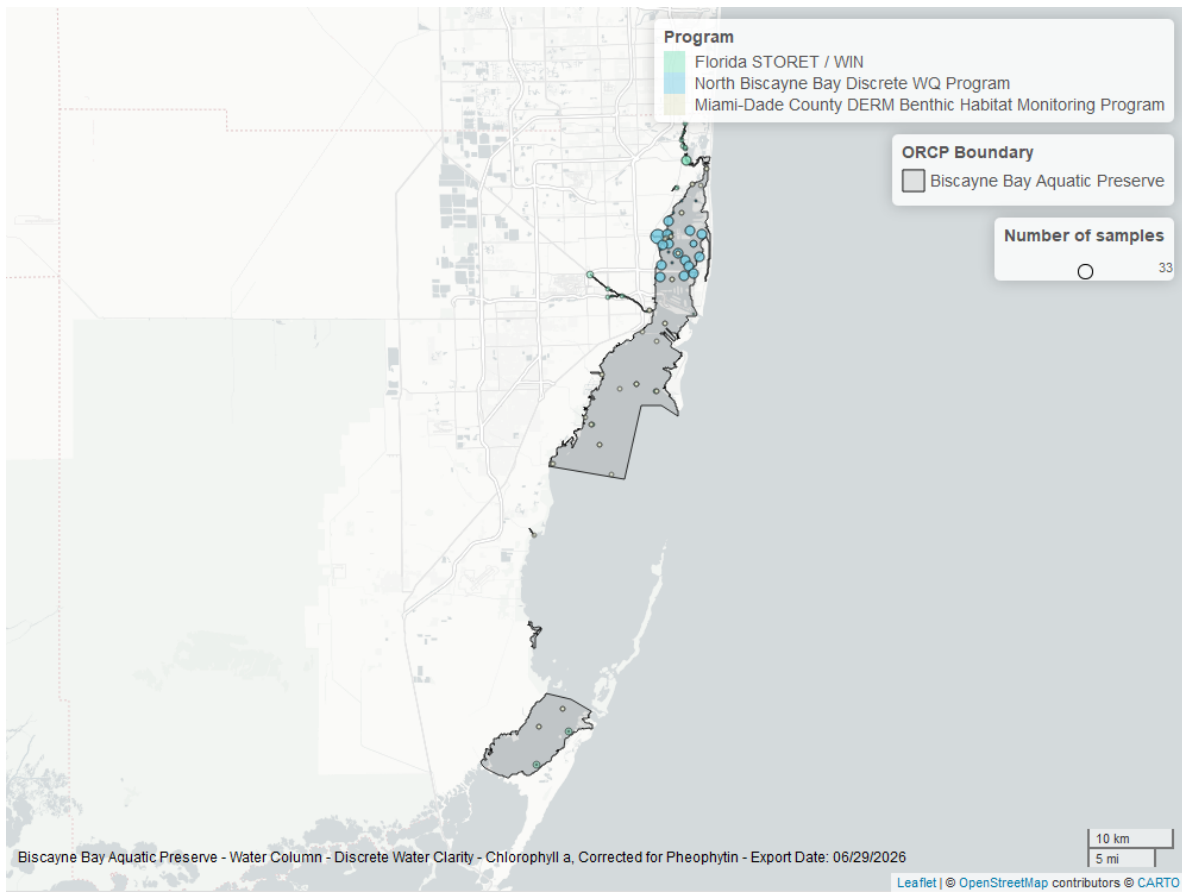


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

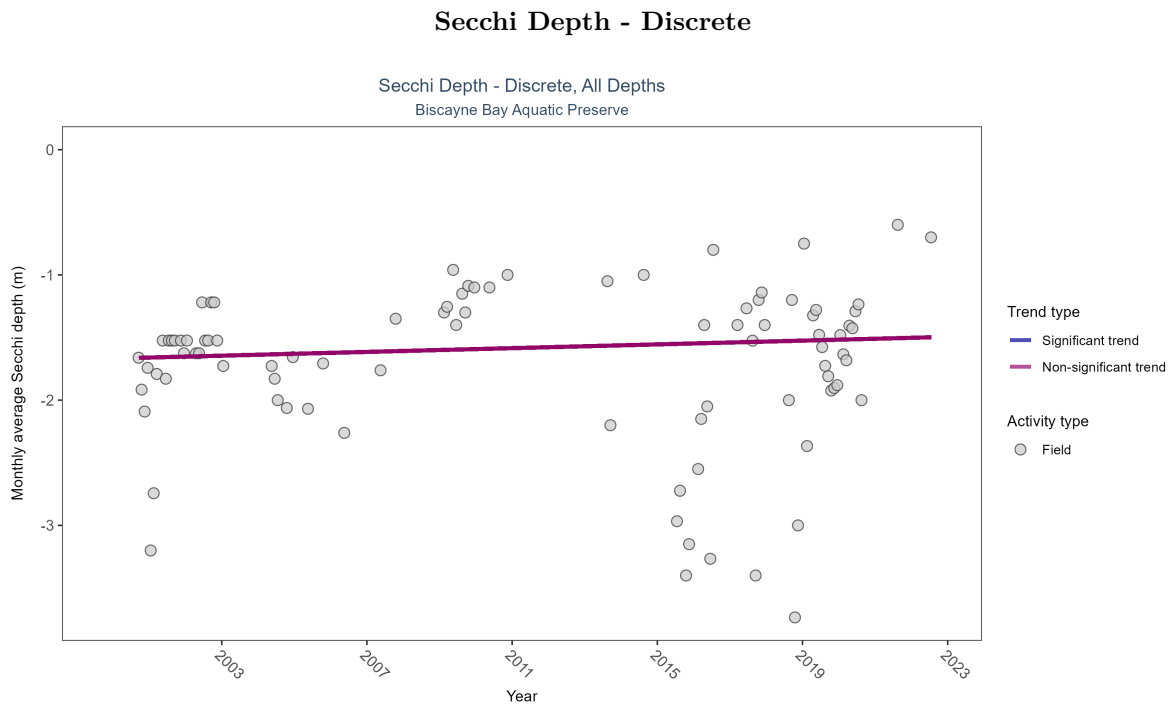


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Secchi depth showed no detectable trend between 2000 and 2022.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	694	20	2000 - 2022	-1.52402	0.09752	-1.66734	0.00751	0.5251

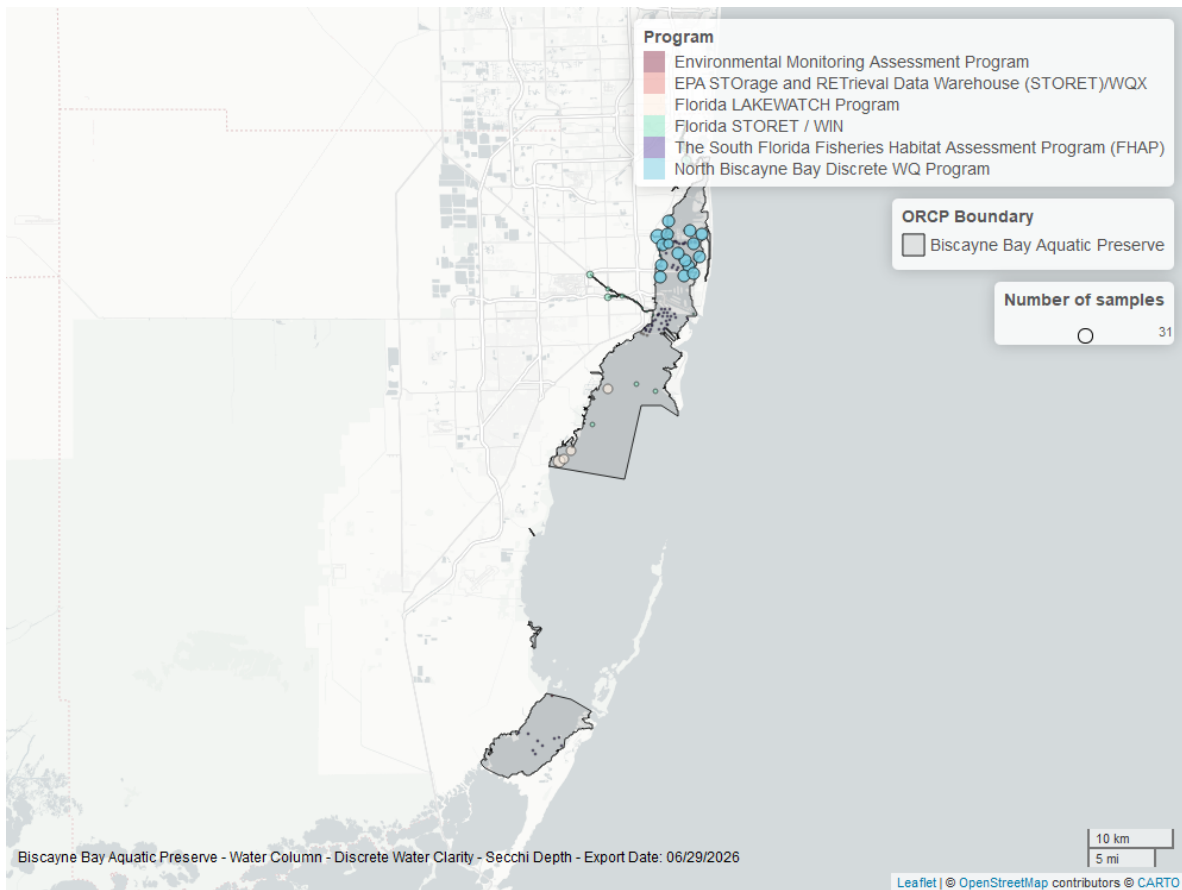


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

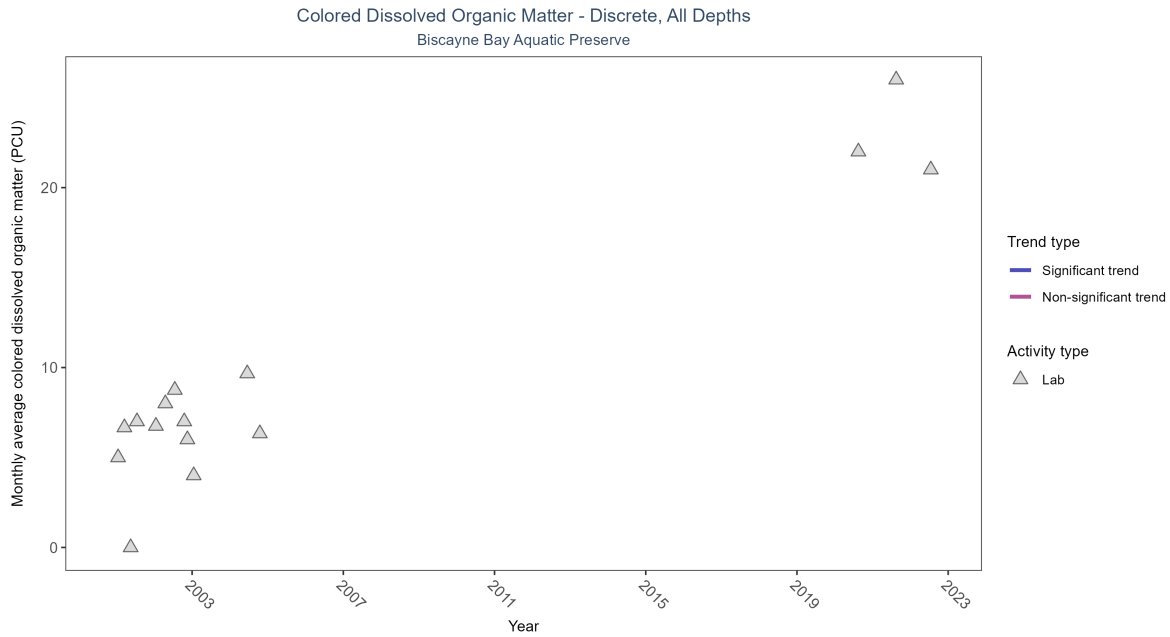


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	38	7	2001 - 2022	7	-	-	-	-

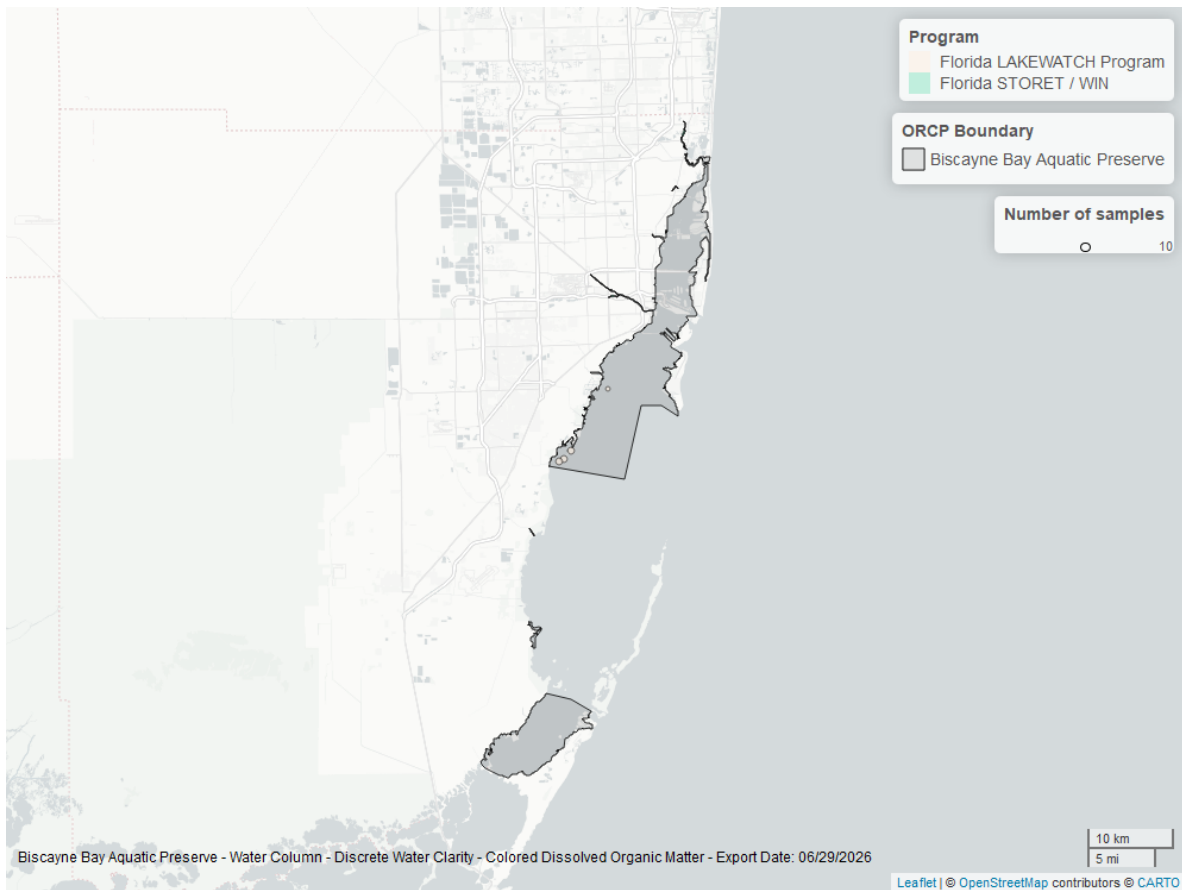


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.